

Lecture 10: Introduction to Data Visualization Continued

Heather Mattie, PhD

October 2, 2025

Recipe of the Day!

Double Crust Chicken Pot Pie

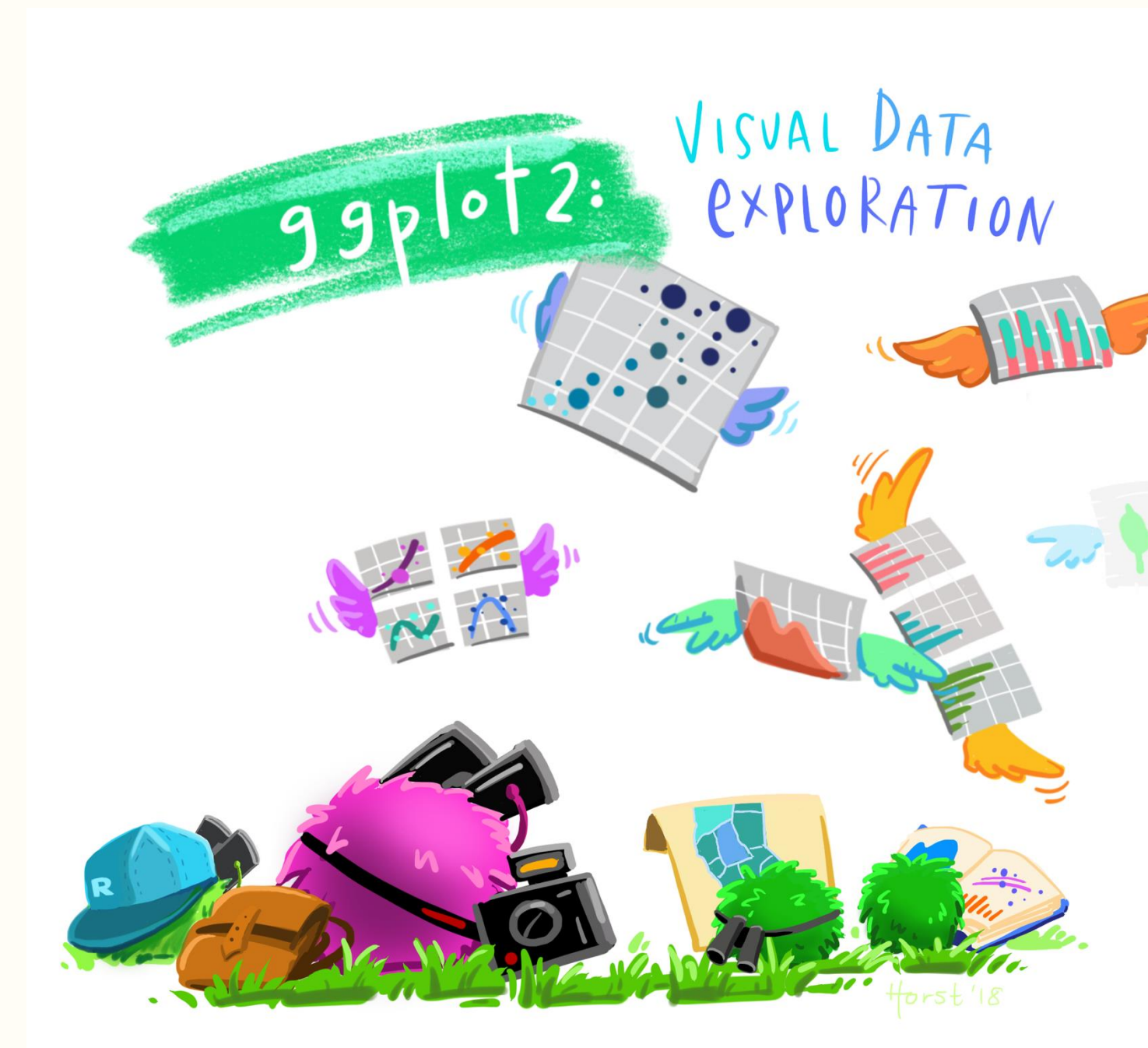


Happy October!



Announcements

- Homework #2 due Friday, October 10 by 11:59pm
- Lab will be held this week



Coding question of the day

Using the `gapminder` dataset and `ggplot2`, for the year 1986, plot `fertility` on the x-axis and `life_expectancy` on the y-axis. Color the points according to `continent`, and make the size of the points correspond to the `population` of the country.

Update the x and y axis labels to make them more informative. Hint: the arguments `color` and `size` will be useful inside of `ggplot`. Note: you will most likely get ugly legends - don't worry about that right now.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data(gapminder)
```


Introduction to Data Visualization



Data Visualization Roadmap

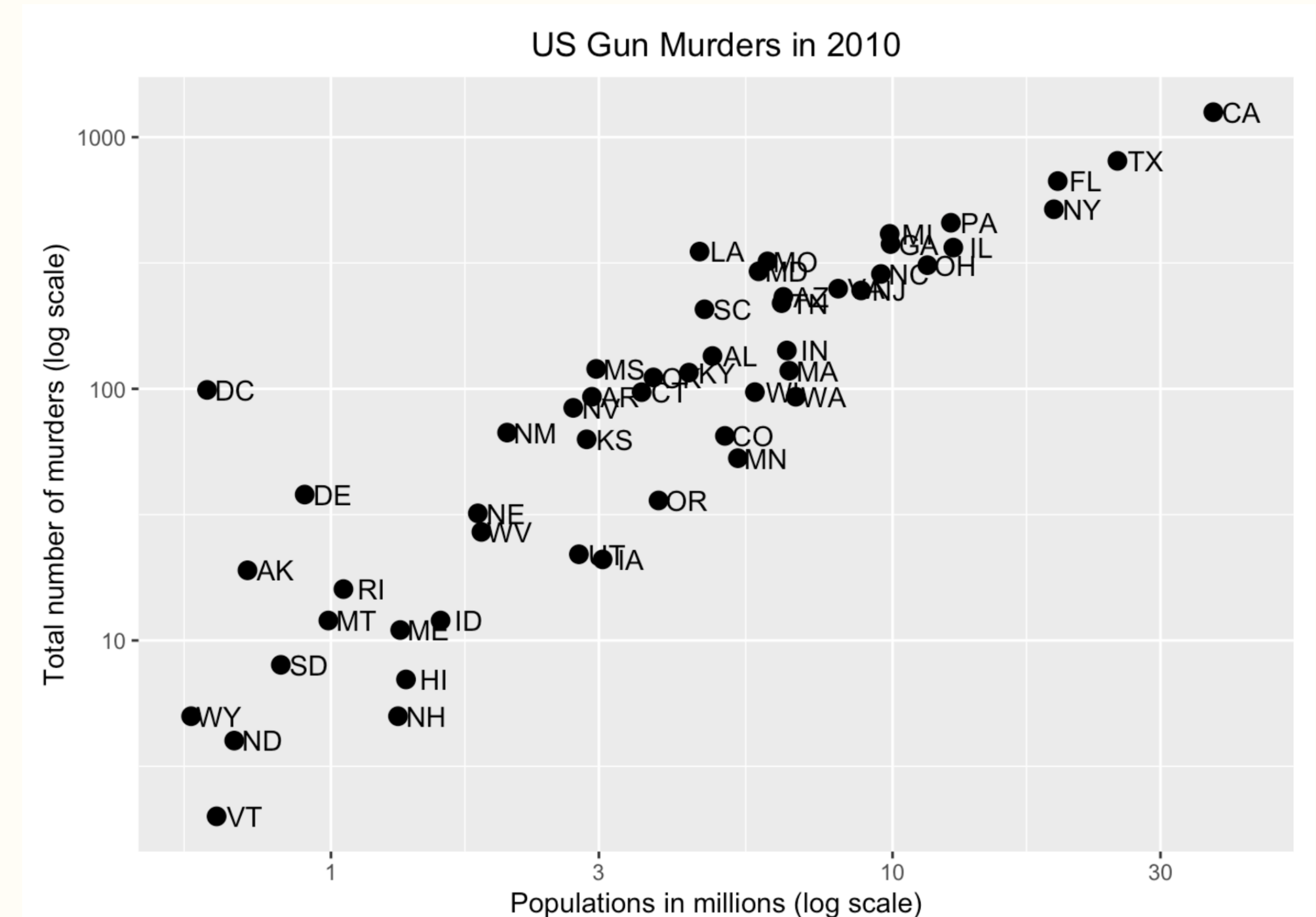
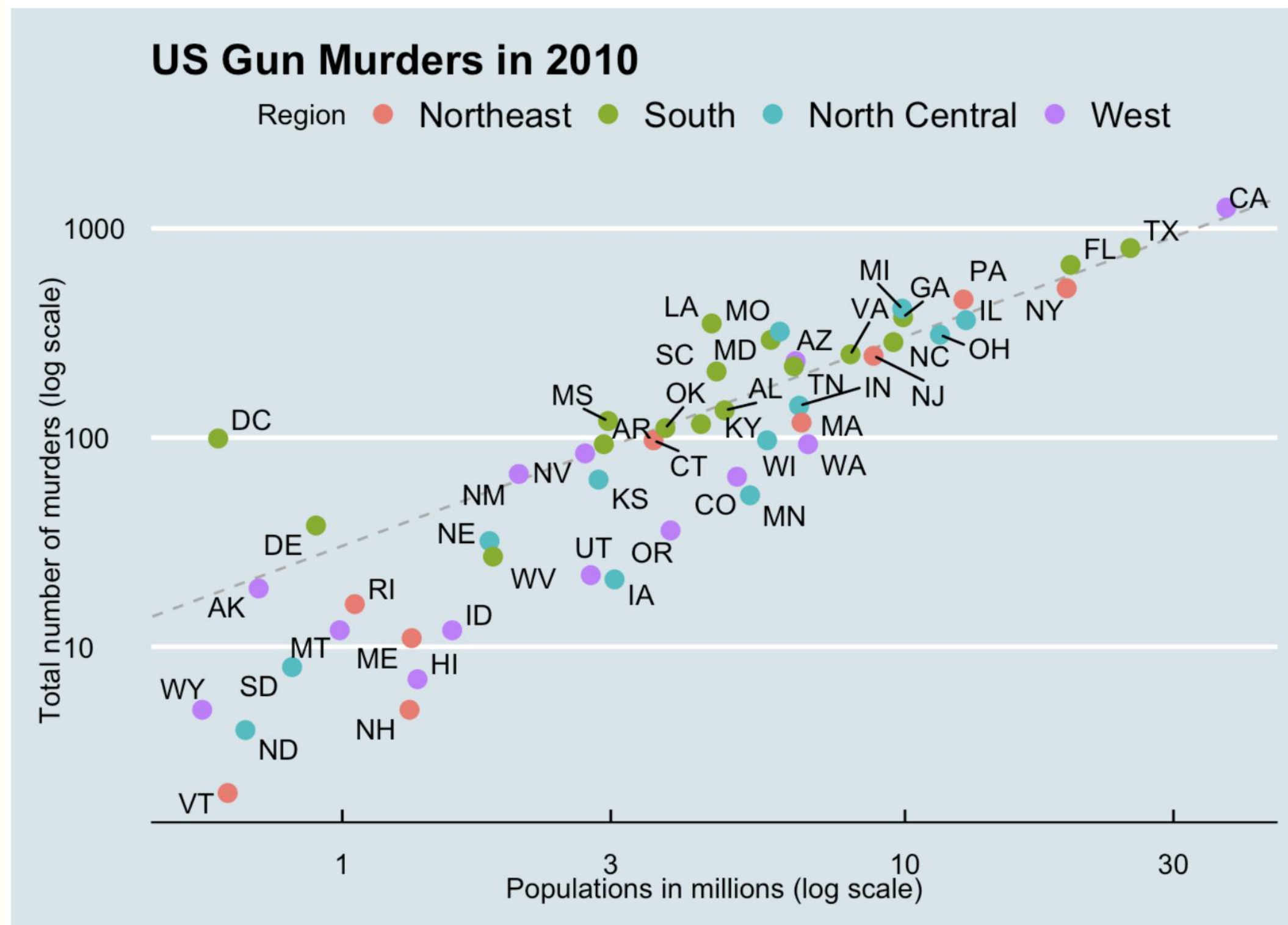
1. Introduction to `ggplot2`
2. Fine-tuning visualizations
3. Data visualization principles
4. Case study: vaccines

Introduction to data visualization

Fine-tuning visualizations

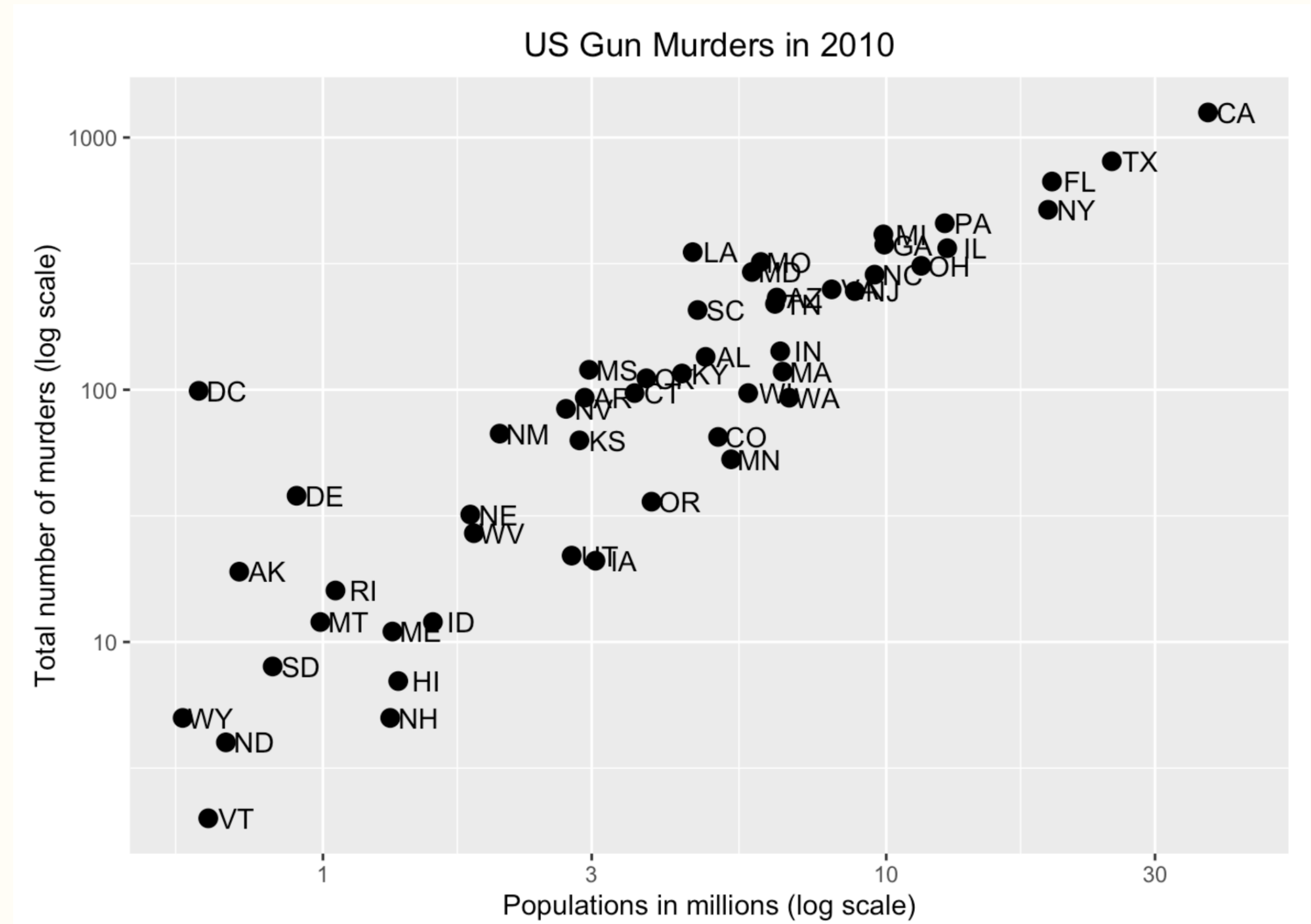
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Progress check



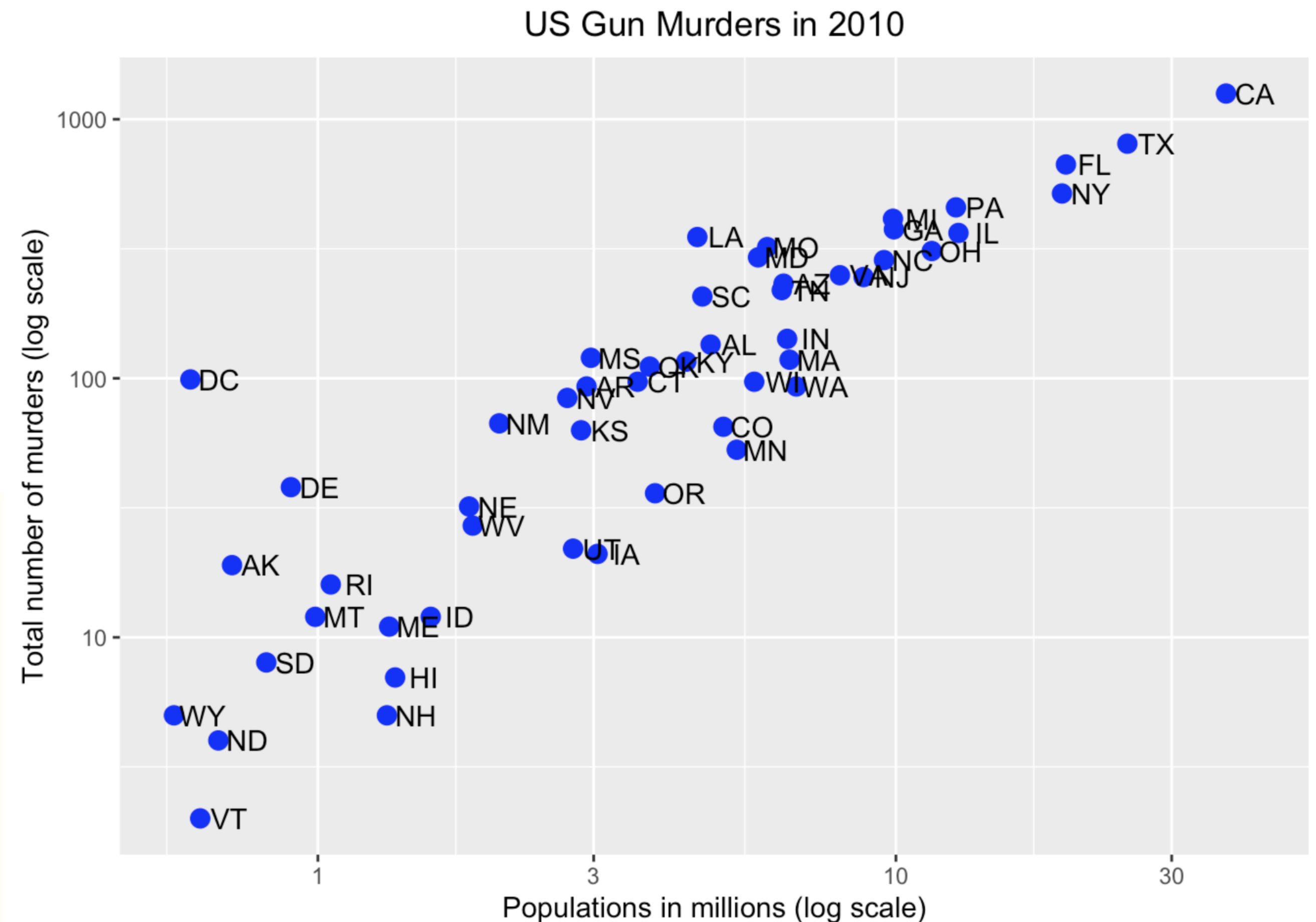
Progress check

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



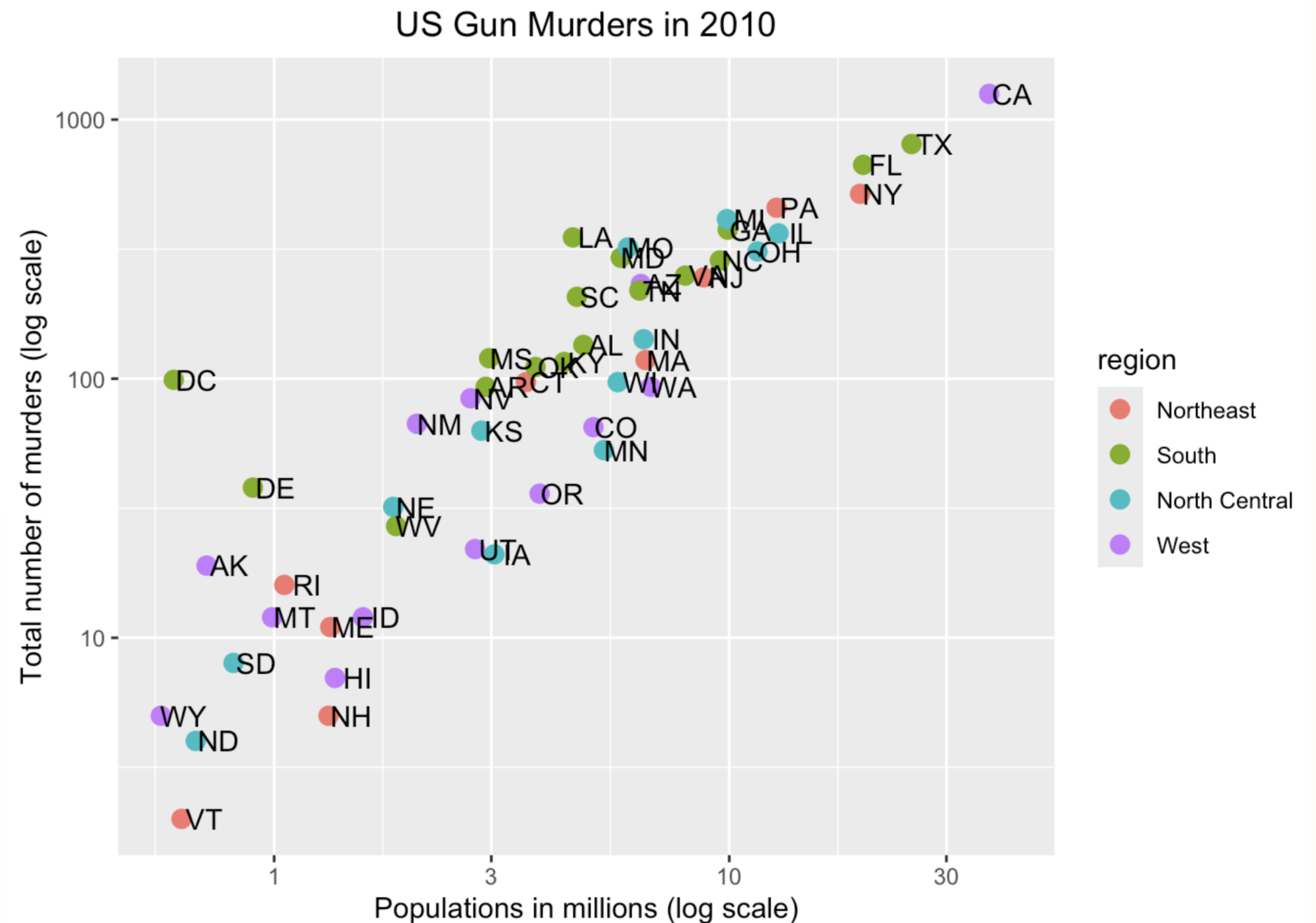
Categories and colors

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(size = 3, color = "blue") +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



Categories and colors

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(aes(color = region), size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```

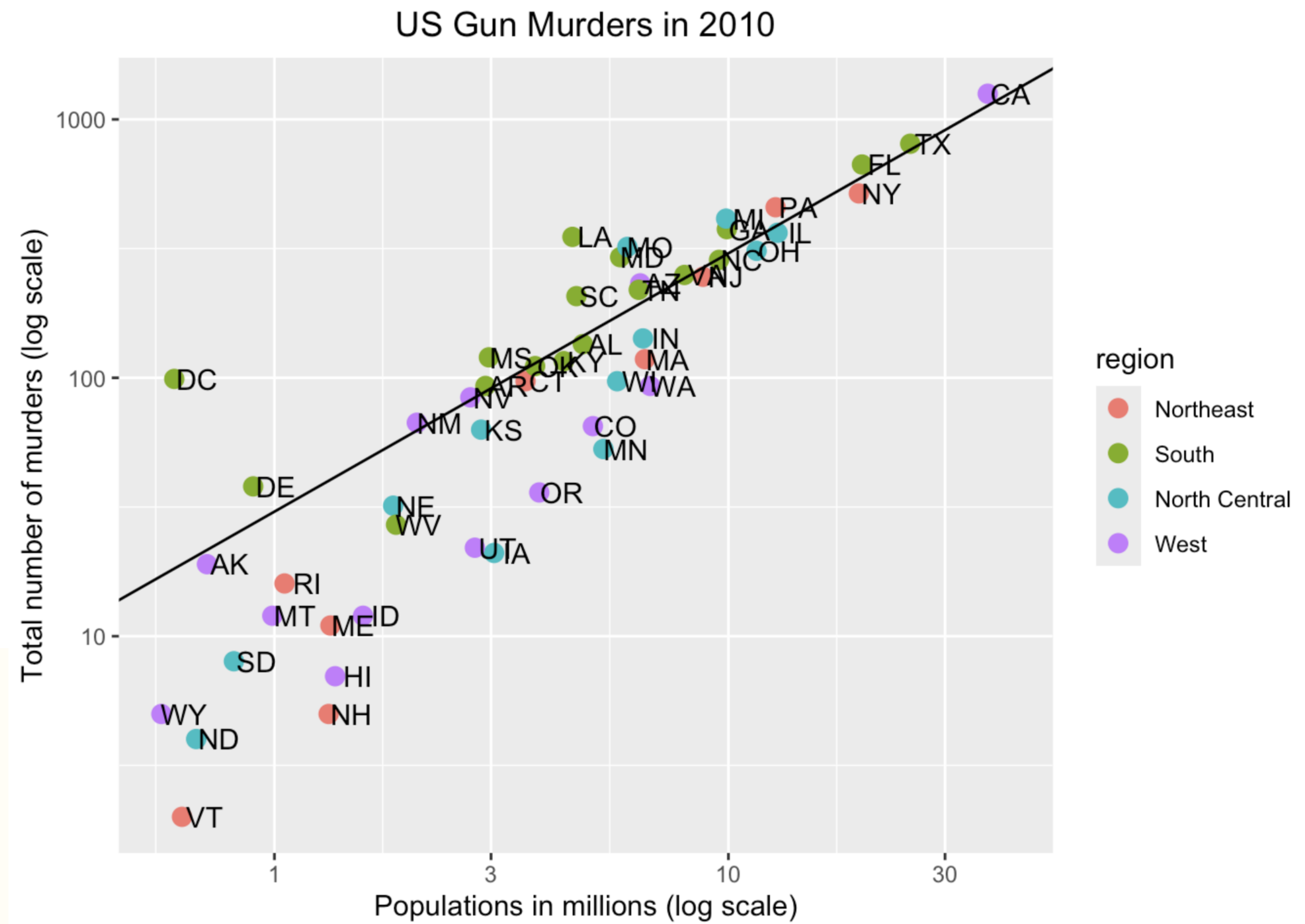


Adding a line

```
r <- murders %>%  
  summarize(murder_rate = sum(total) / sum(population) * 10^6) %>% . $murder_rate  
r
```

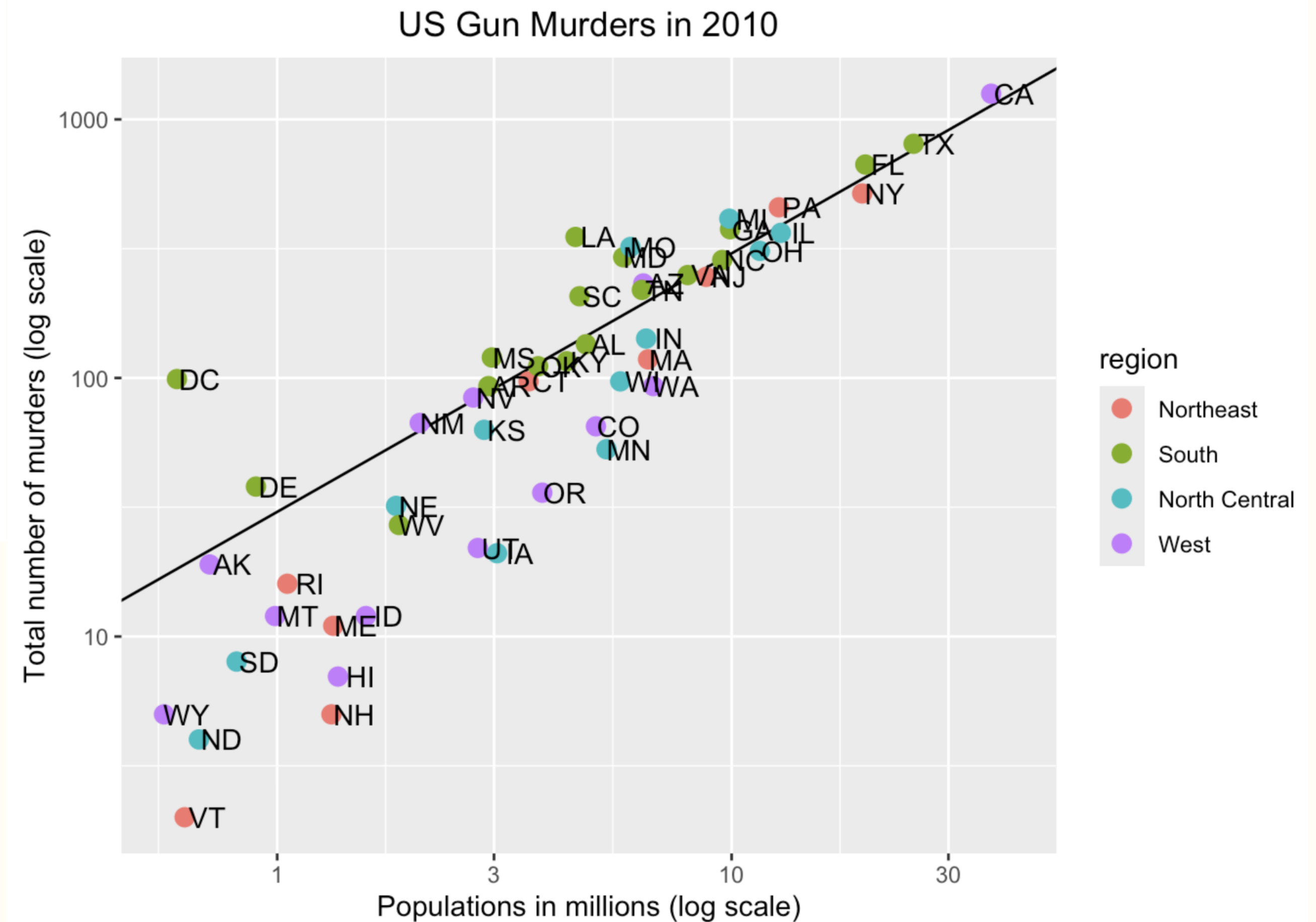
```
## [1] 30.34555
```

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_point(aes(color = region), size = 3) +  
  geom_abline(intercept = log10(r)) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



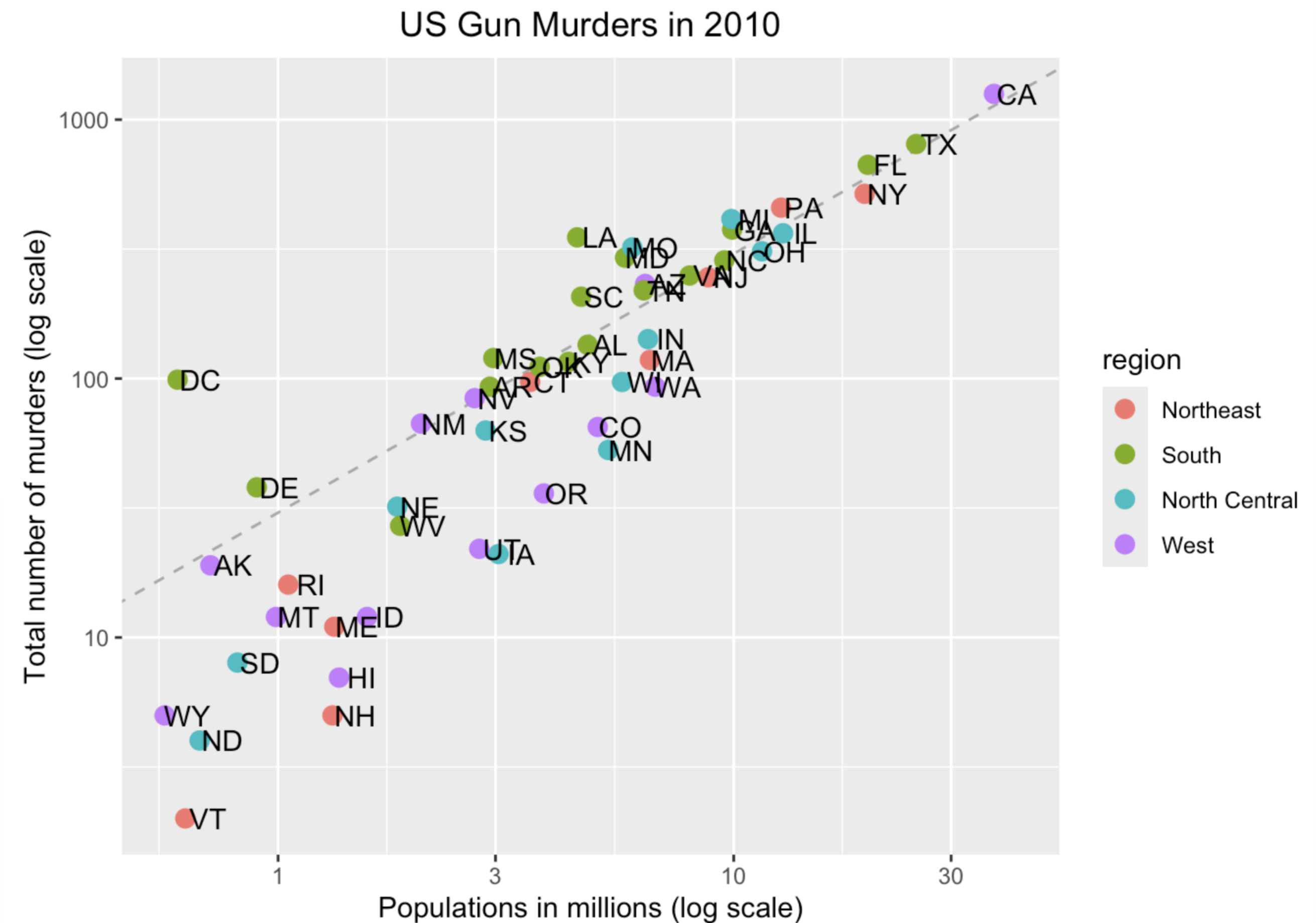
Adding a line

```
murders %>%  
  ggplot(aes(x = population/10^6, y = total, label = abb)) +  
  geom_abline(intercept = log10(r)) +  
  geom_point(aes(color = region), size = 3) +  
  geom_text(nudge_x = 0.05) +  
  scale_x_log10() +  
  scale_y_log10() +  
  xlab("Populations in millions (log scale)") +  
  ylab("Total number of murders (log scale)") +  
  ggtitle("US Gun Murders in 2010") +  
  theme(plot.title = element_text(hjust = 0.5))
```



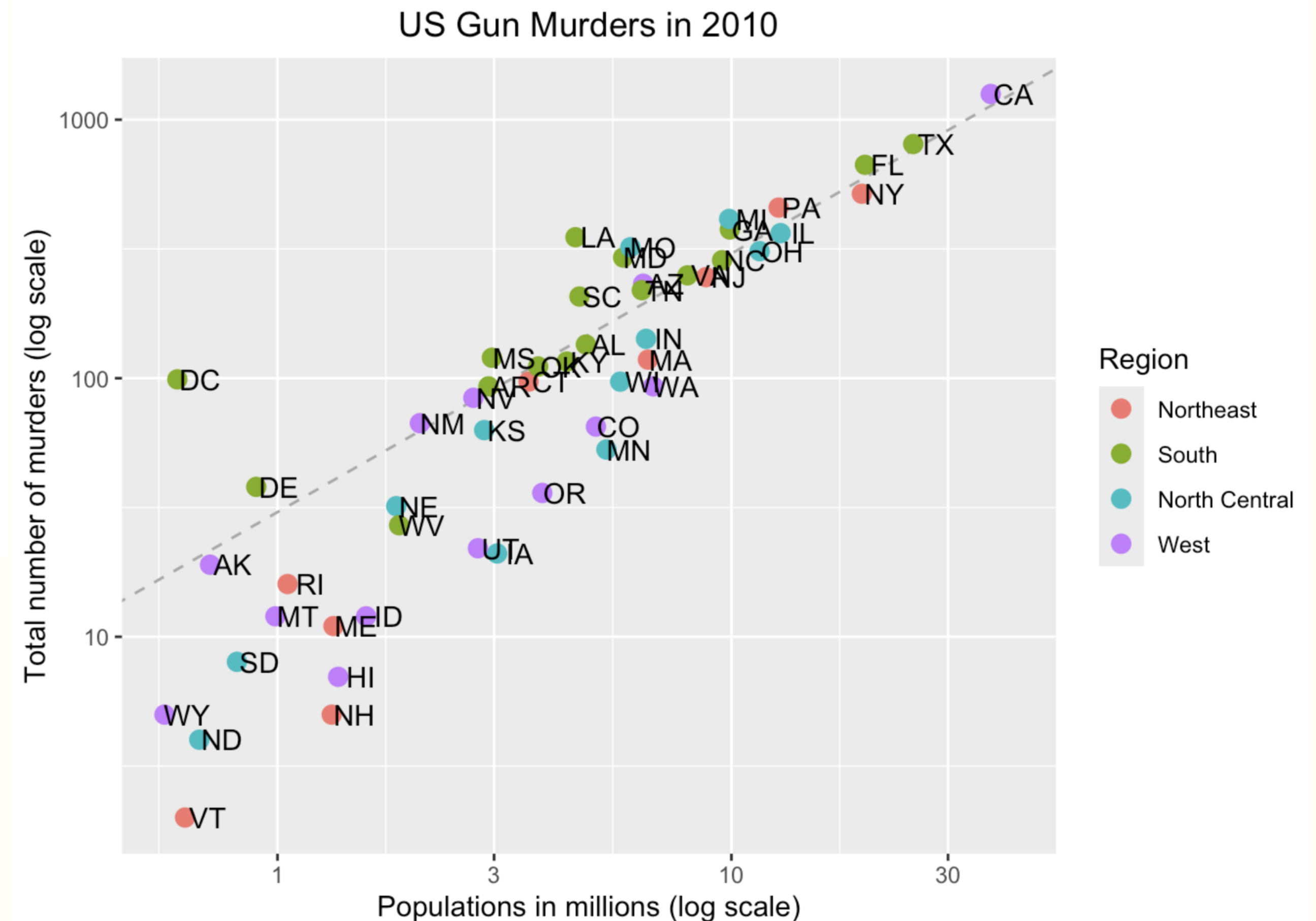
Adding a line

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5))
```



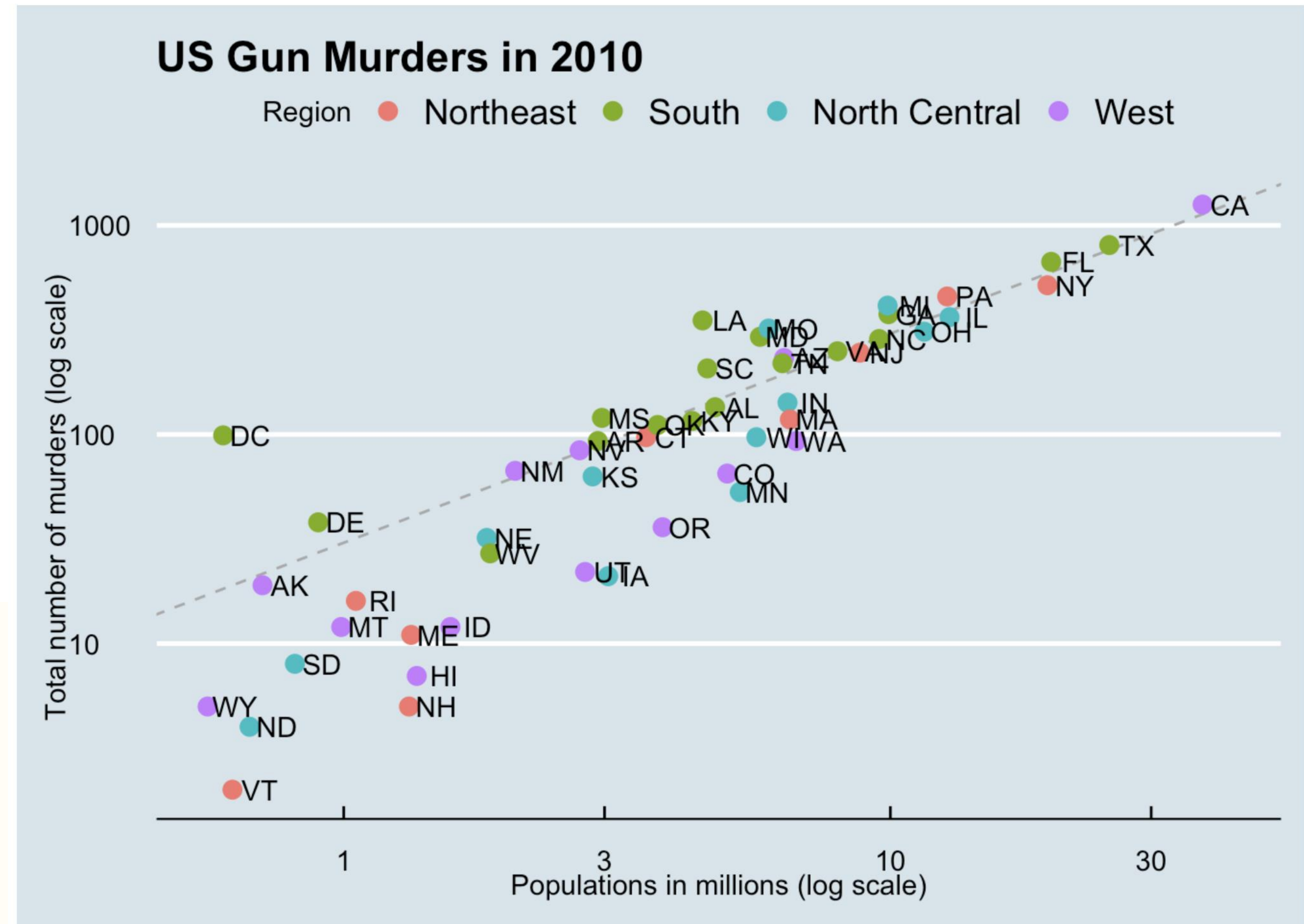
Legend title

```
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_color_discrete(name = "Region")
```



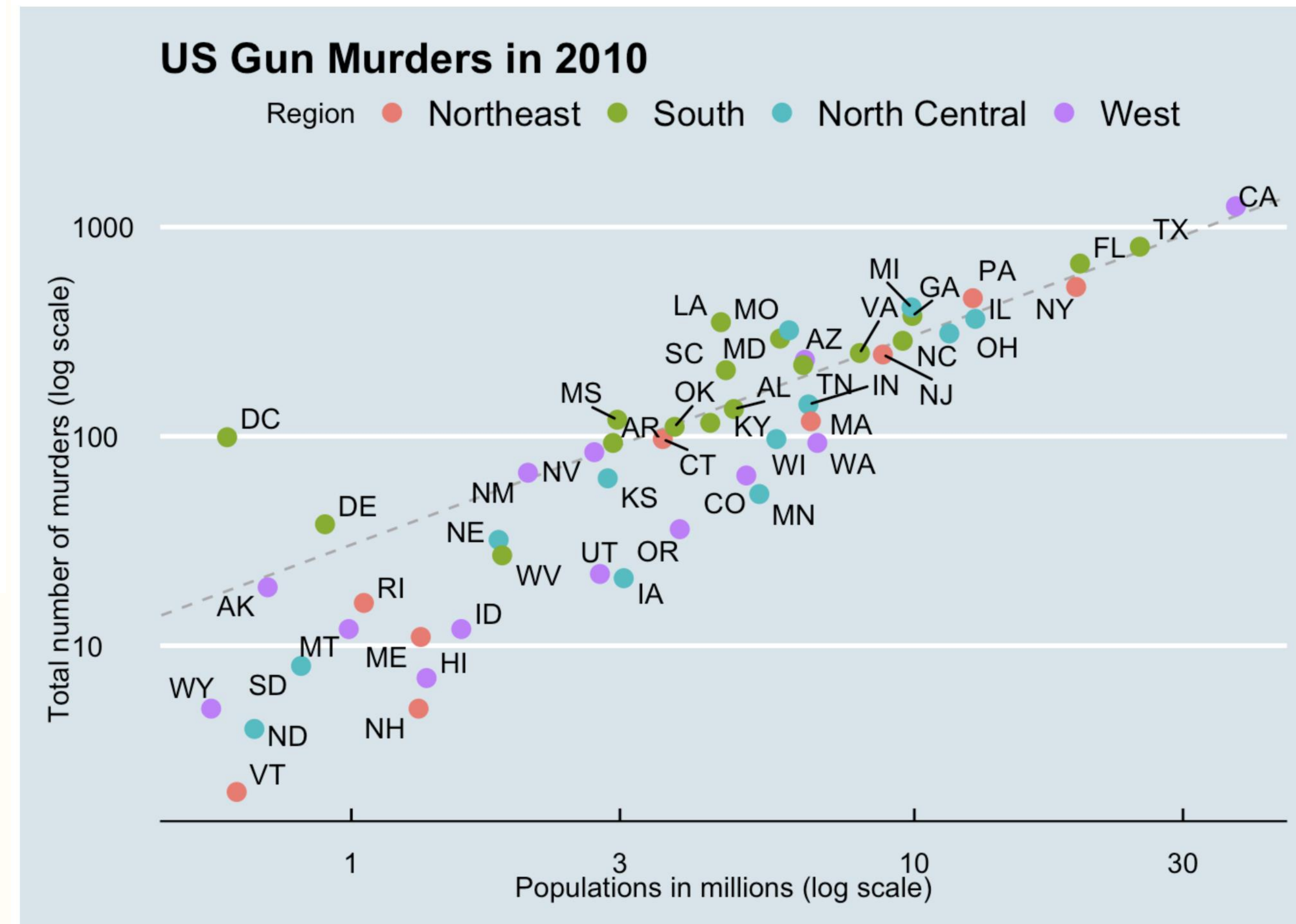
ggthemes

```
library(ggthemes)
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text(nudge_x = 0.05) +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



ggrepel

```
library(ggrepel)
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text_repel() +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



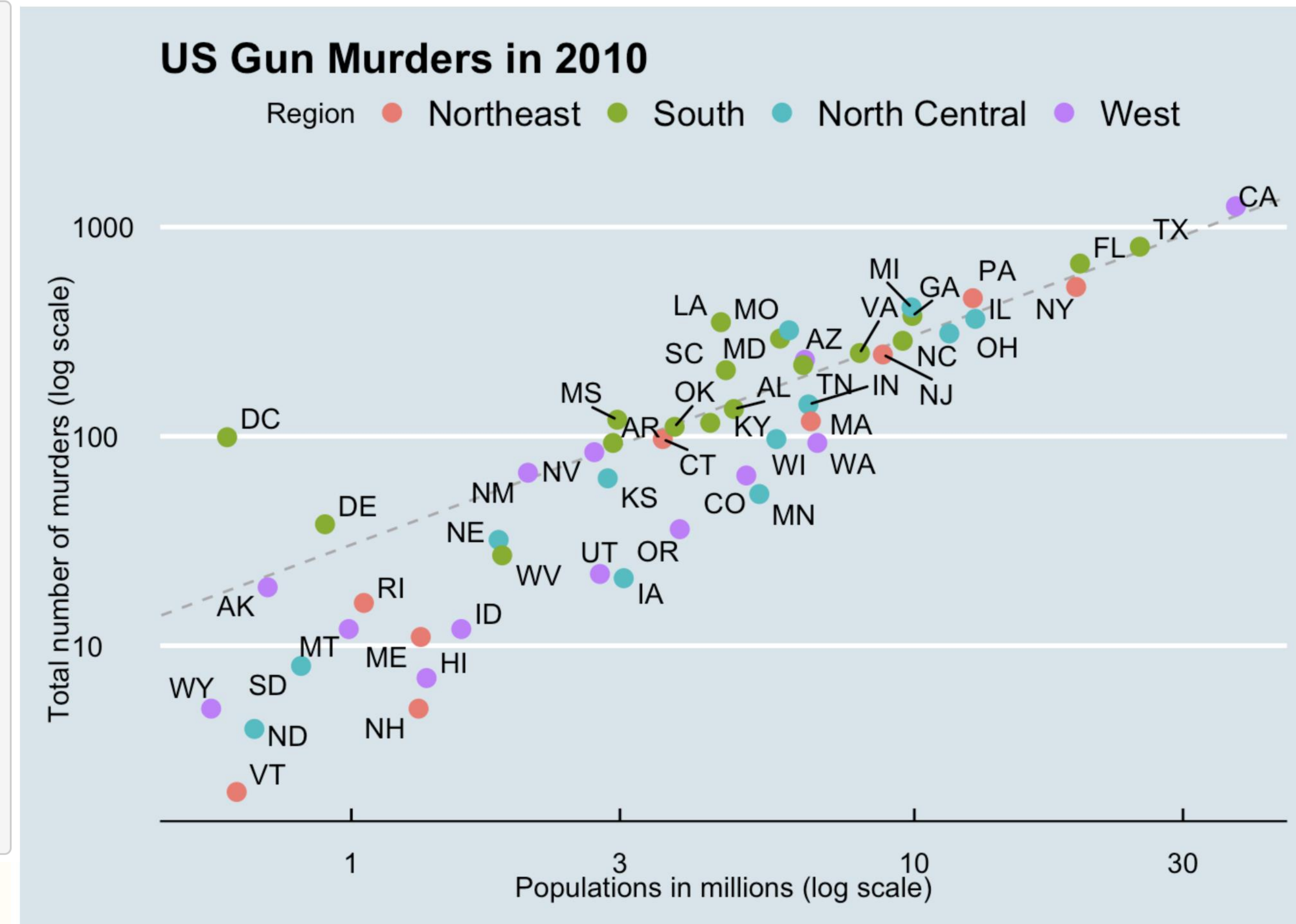
Putting it all together

```
library(tidyverse)
library(dslabs)
library(ggthemes)
library(ggrepel)

## Load the data
data(murders)

## First define the slope of the line
r <- murders %>%
  summarize(murder_rate= sum(total) / sum(population) * 10^6) %>% .$murder_rate

## Now make the plot
murders %>%
  ggplot(aes(x = population/10^6, y = total, label = abb)) +
  geom_abline(intercept = log10(r), lty = 2, color = "darkgrey") +
  geom_point(aes(color = region), size = 3) +
  geom_text_repel() +
  scale_x_log10() +
  scale_y_log10() +
  xlab("Populations in millions (log scale)") +
  ylab("Total number of murders (log scale)") +
  ggtitle("US Gun Murders in 2010") +
  scale_color_discrete(name = "Region") +
  theme_economist()
```



Summary

- Data visualizations created with **ggplot2** are highly customizable
- Plot appearance can be customized using themes, labels, colors, and scales

Introduction to data visualization

Data visualization principles

Heather Mattie, PhD

Data visualization principles

Data visualization principles are mostly based on research related to how humans detect patterns and make visual comparisons

The preferred approaches are those that best fit the way our **brains process visual information**

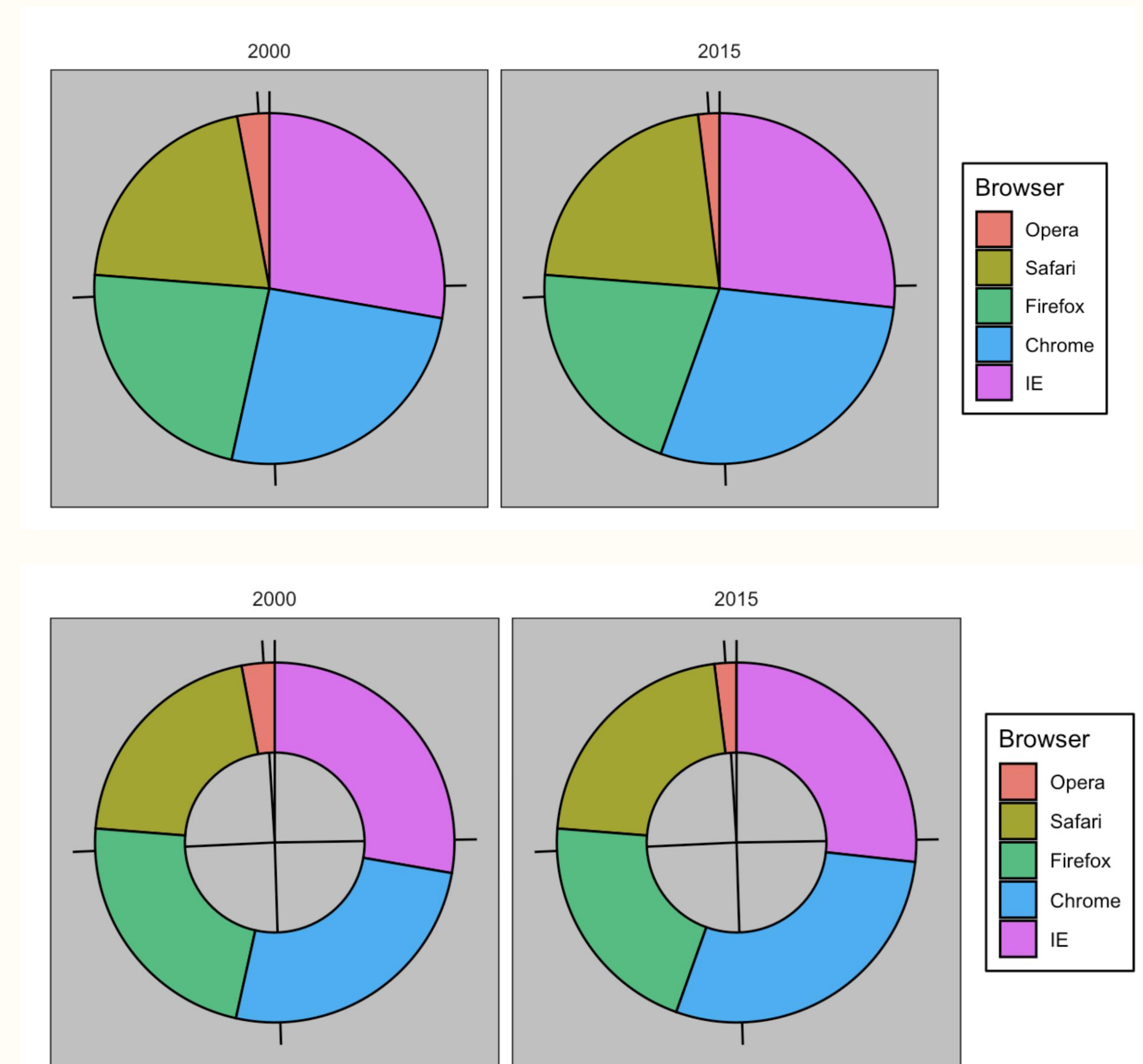
There are several components to consider when creating effective data visualizations

- Position
- Area
- Aligned lengths
- Brightness
- Angles
- Color hue

Encoding data with visual cues

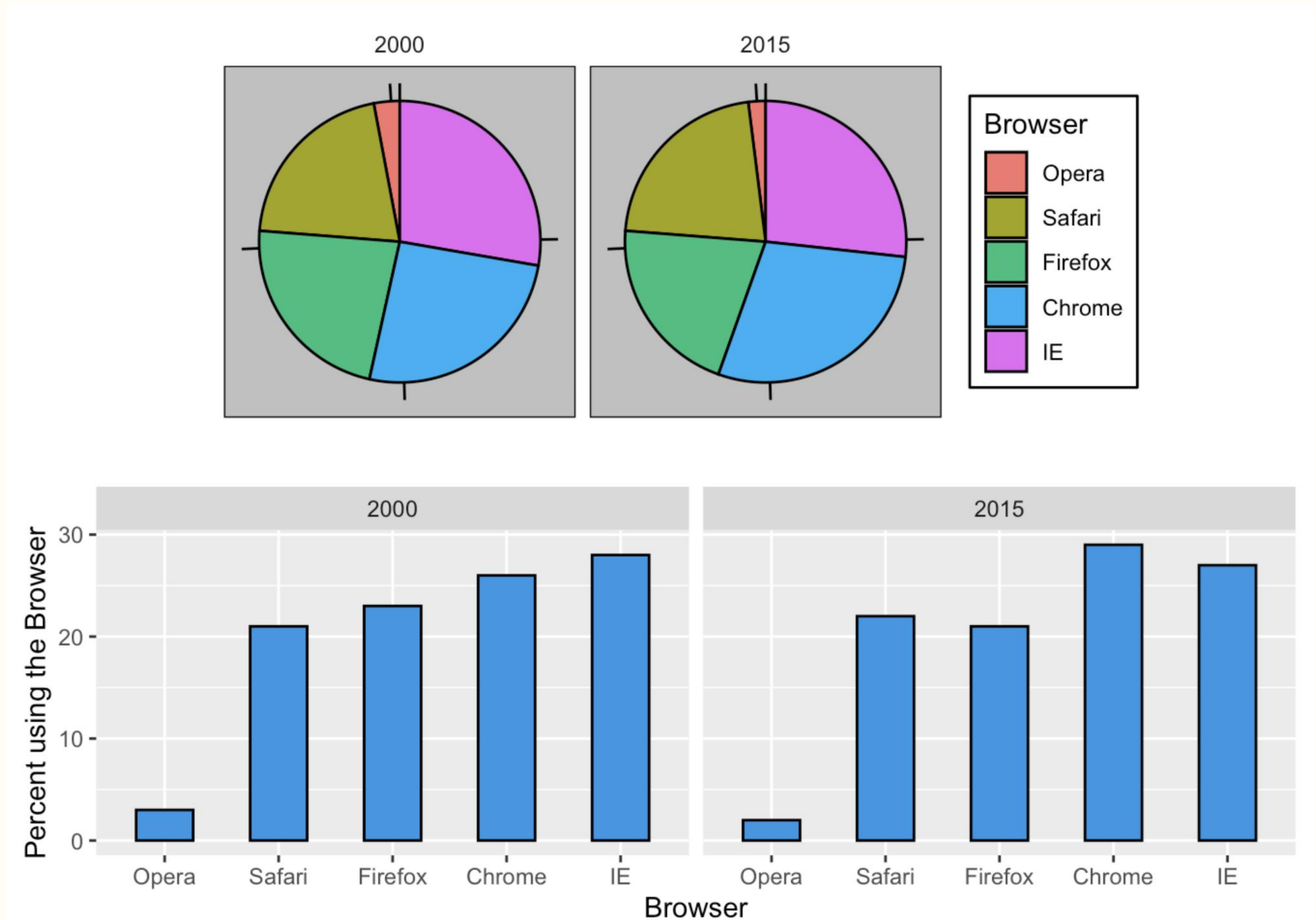
A widely used graphical representation of percentages, popularized by Microsoft Excel, is the **pie chart**

- Humans are not good at precisely quantifying **angles** and are even worse when only **area** is available
- It is much better to use length and position, or just a table of values



Encoding data with visual cues

Browser	2000	2015
Opera	3	2
Safari	21	22
Firefox	23	21
Chrome	26	29
IE	28	27



Pie charts

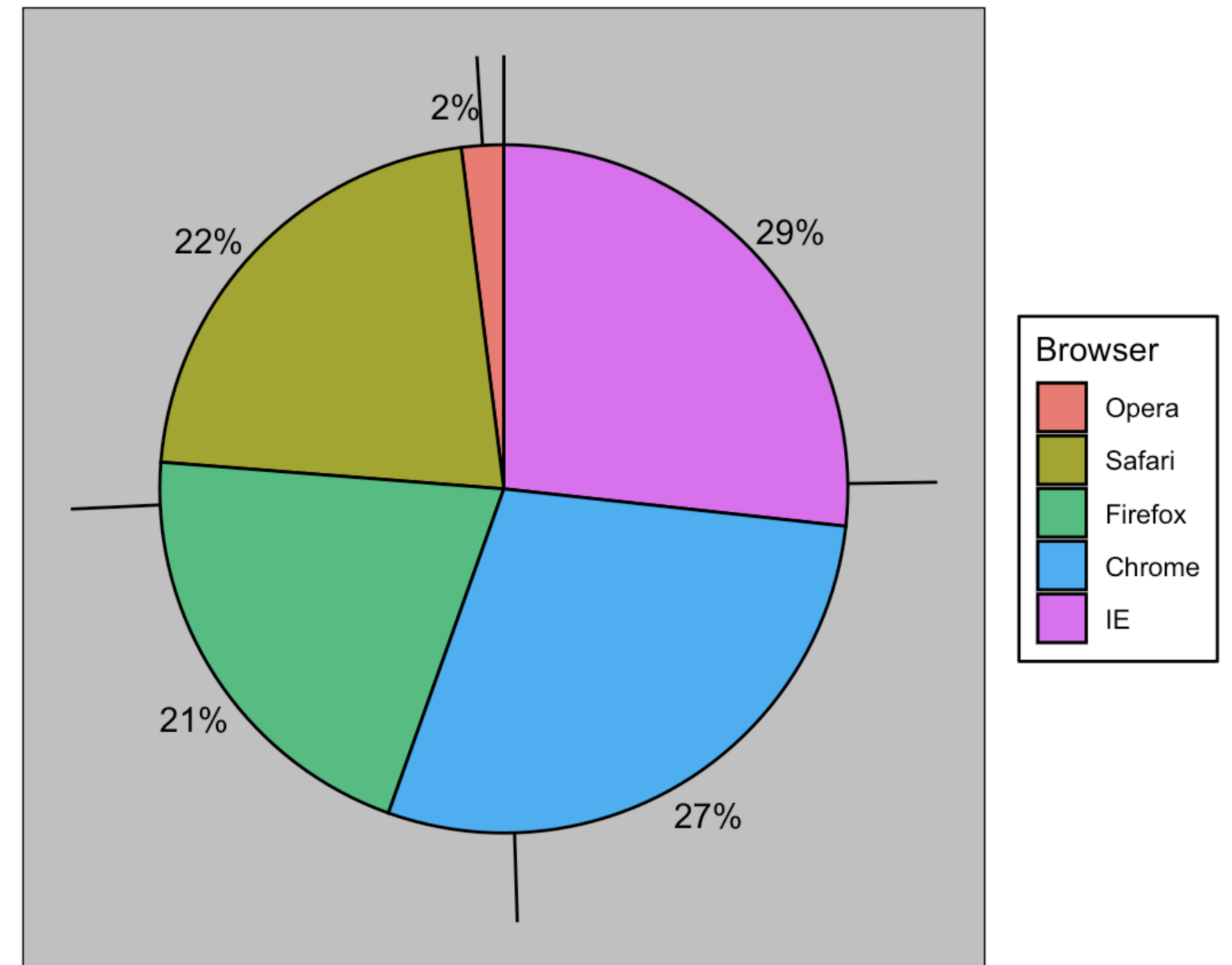
The **pie** R function help file states:

*"Note: Pie charts are a **very bad way** of displaying information. The eye is good at judging linear measures and bad at judging relative areas. A **bar chart** or **dot chart** is a preferable way of displaying this type of data."*

Pie charts

If you must use a pie chart, be sure to include the **percentages as numbers** to avoid having to infer them from the angles or area

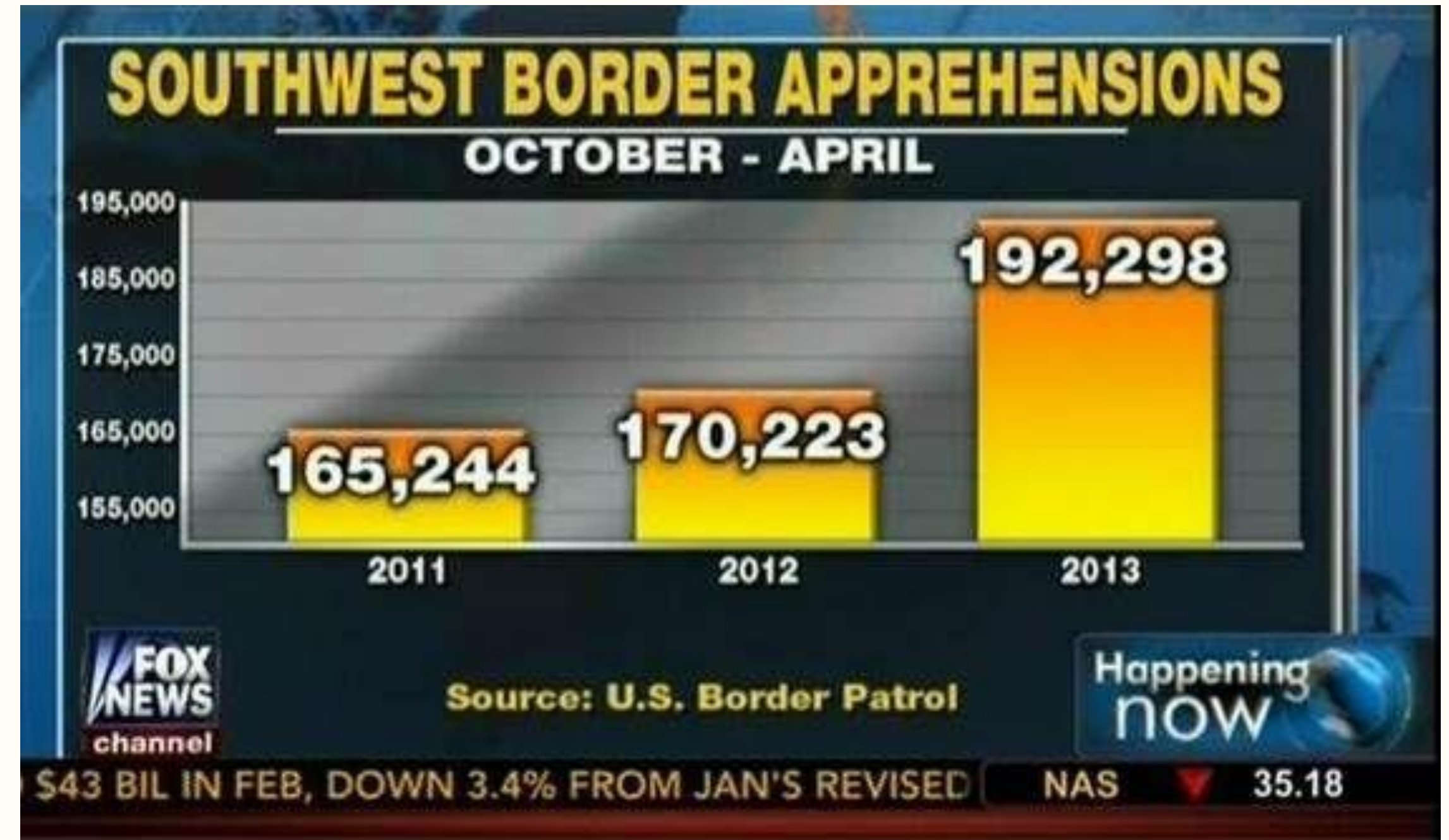
2015



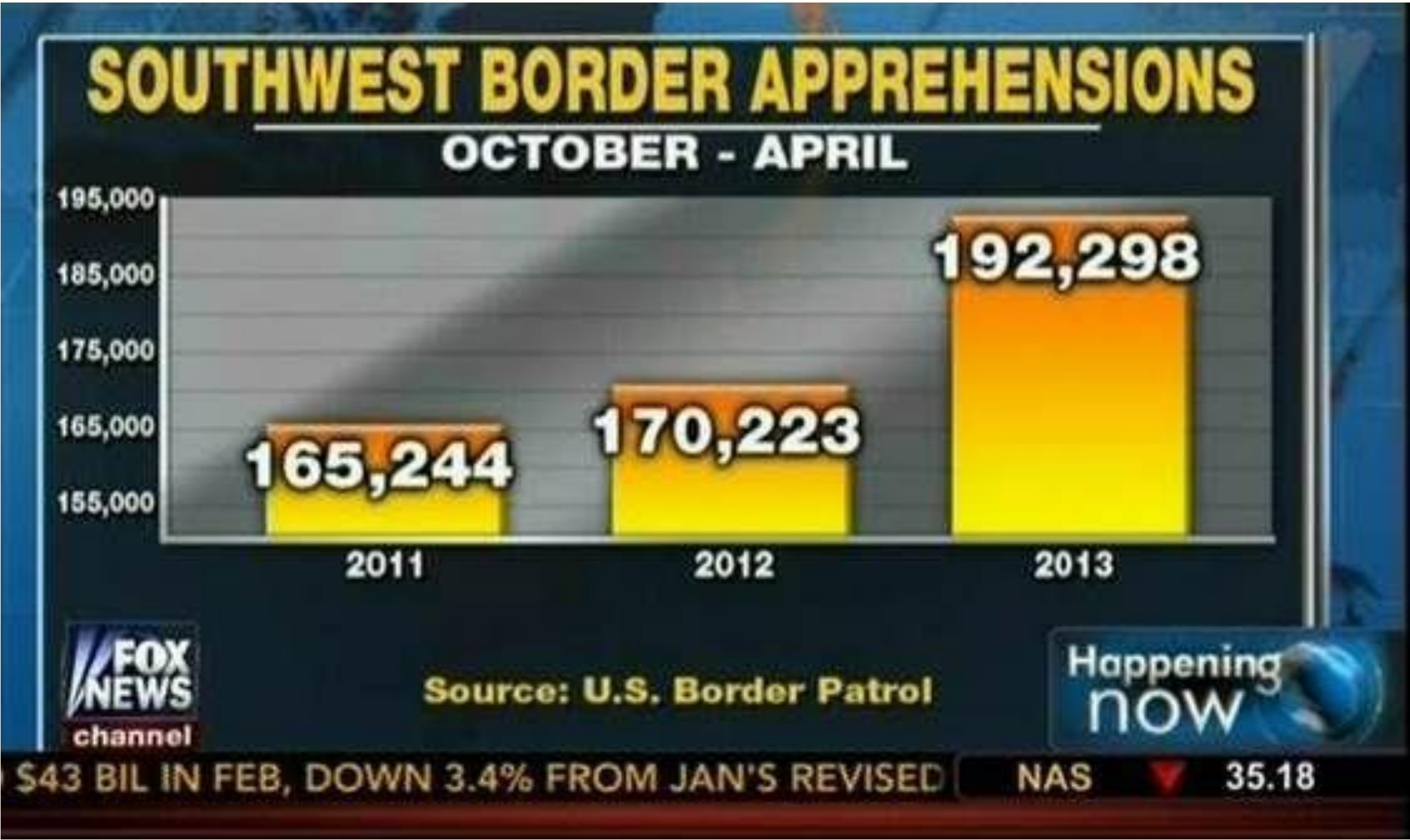
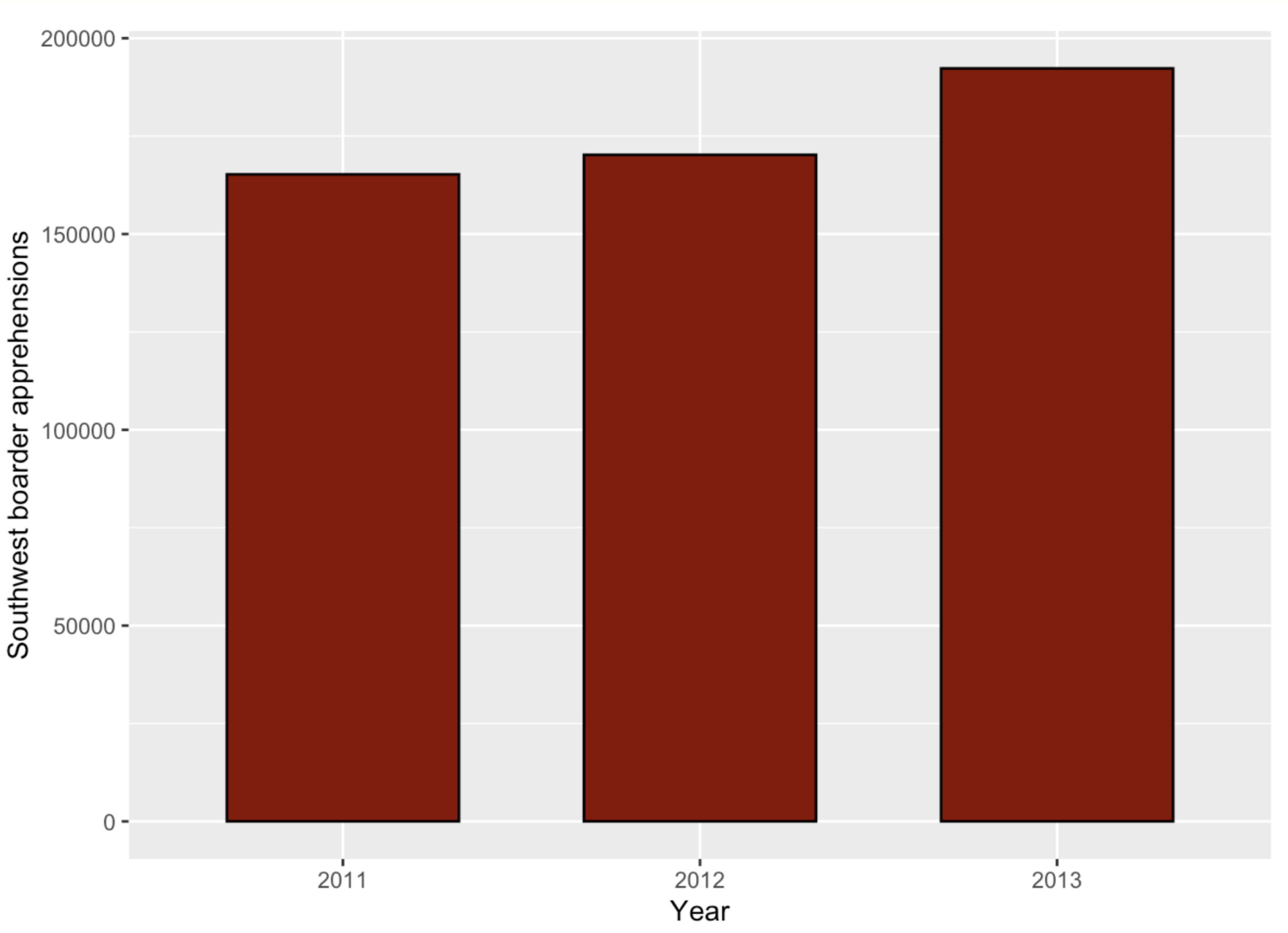
When to include 0

When using barplots, it is **misinformative** to not include 0

- Barplots imply the length is proportional to the quantities being displayed
- The plot here makes it appear that the number of apprehensions has tripled from 2011 to 2013 – but they have only increased by 16%



When to include 0

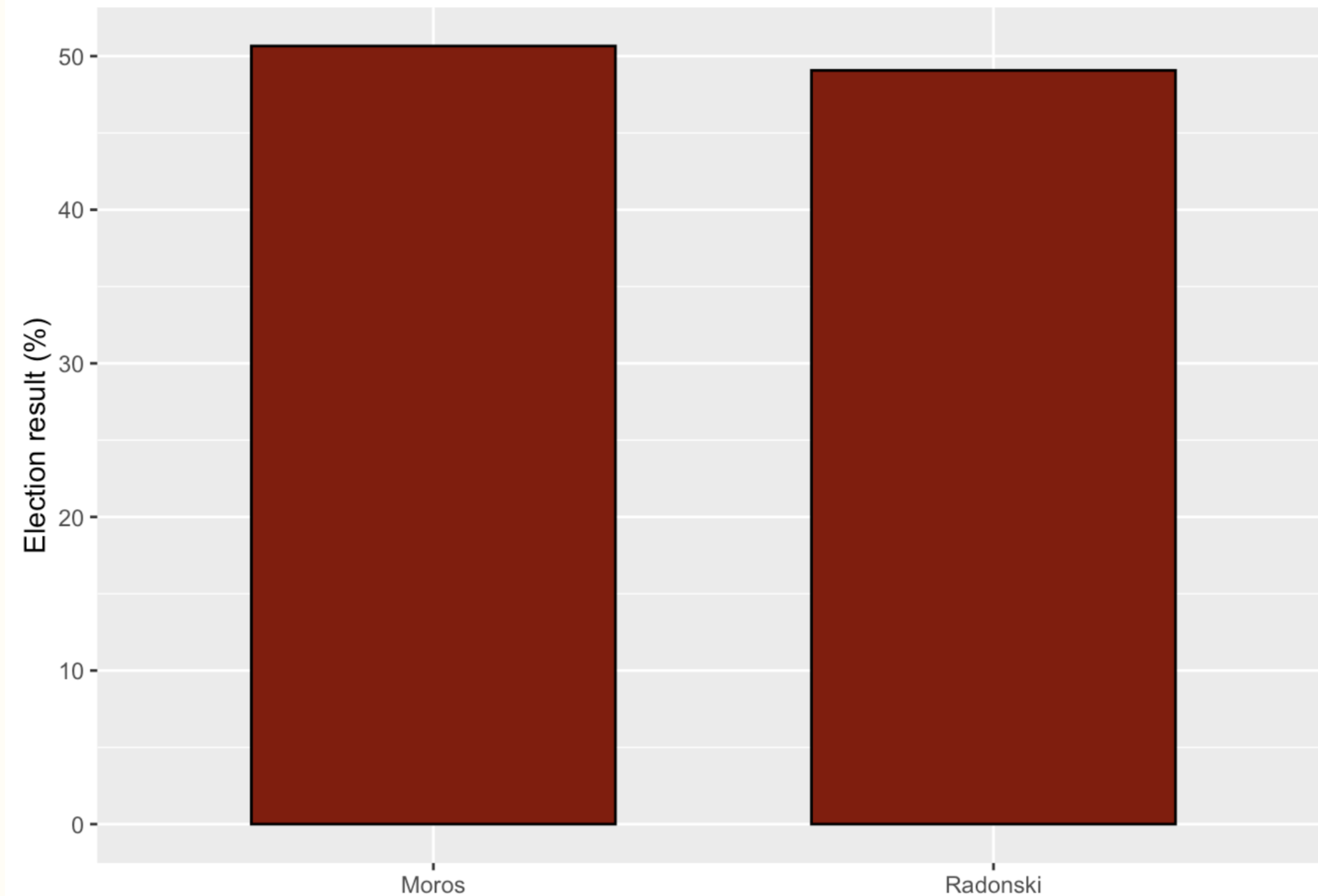


When to include 0

This extreme example makes a very small difference ($<2\%$) look like a 10-fold change

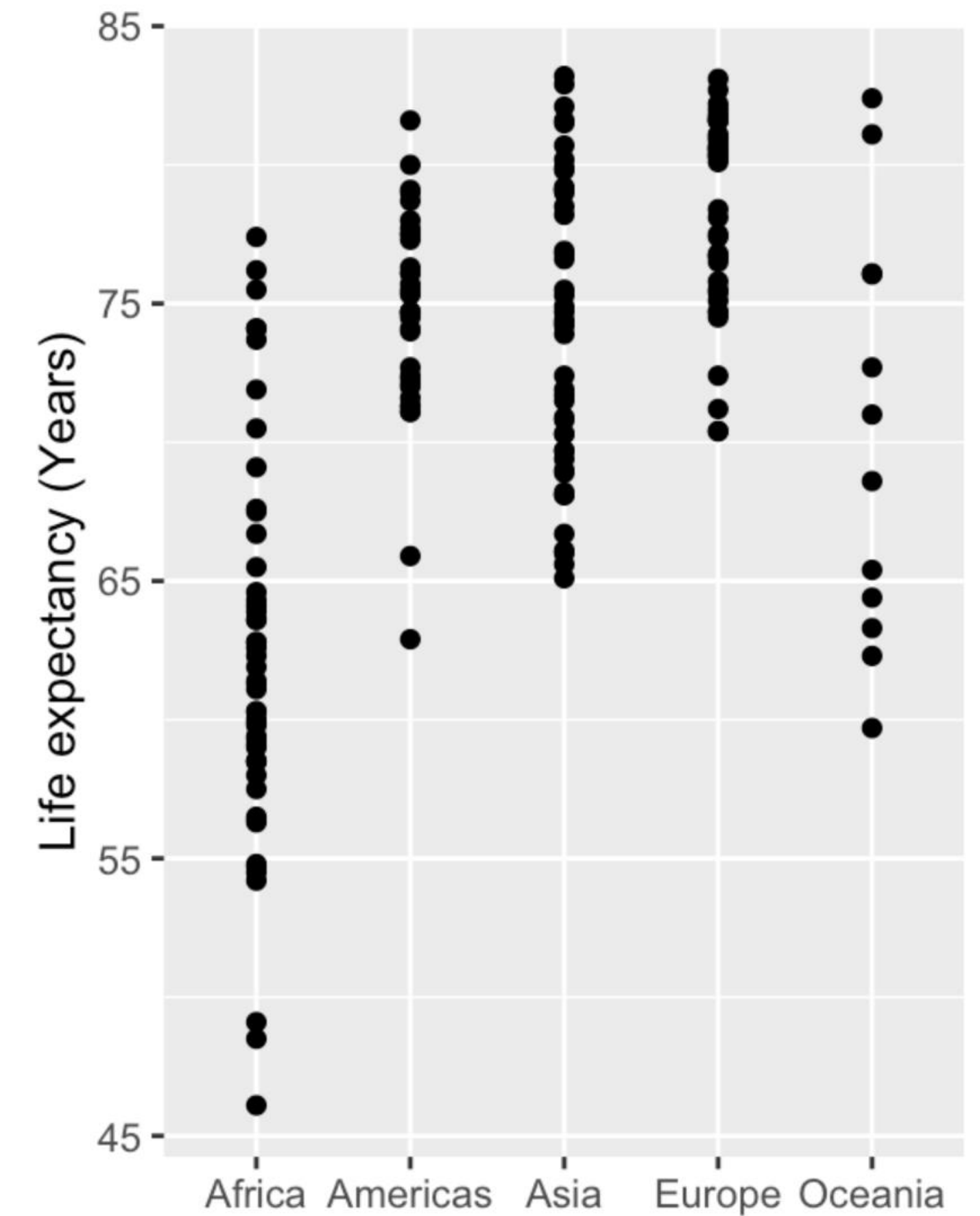
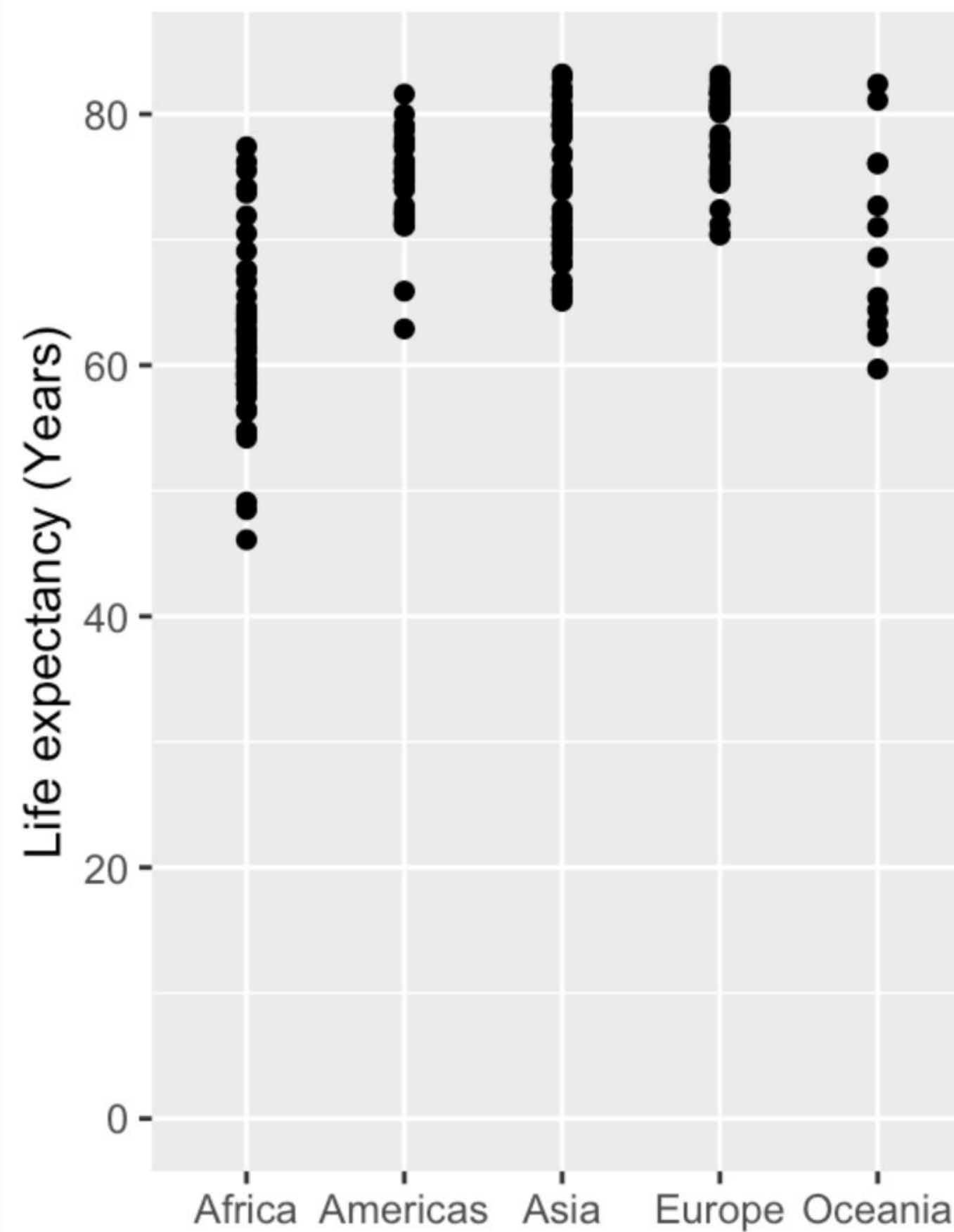


When to include 0



When to include 0

When using position rather than length, it is then **not** necessary to include 0



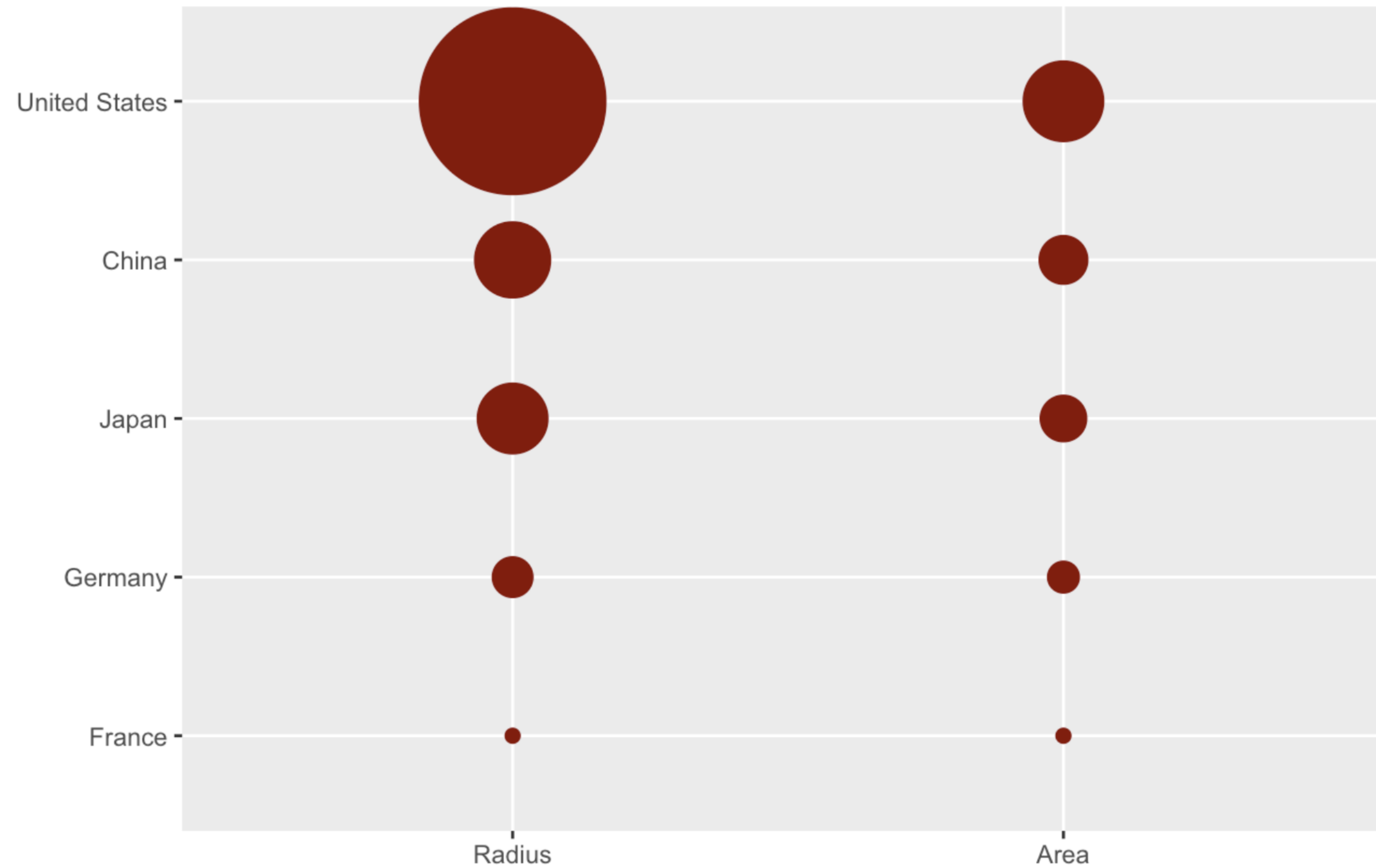
Distortion of quantities

In 2011 this chart was used to compare the US GDP to the GDP of four competing nations

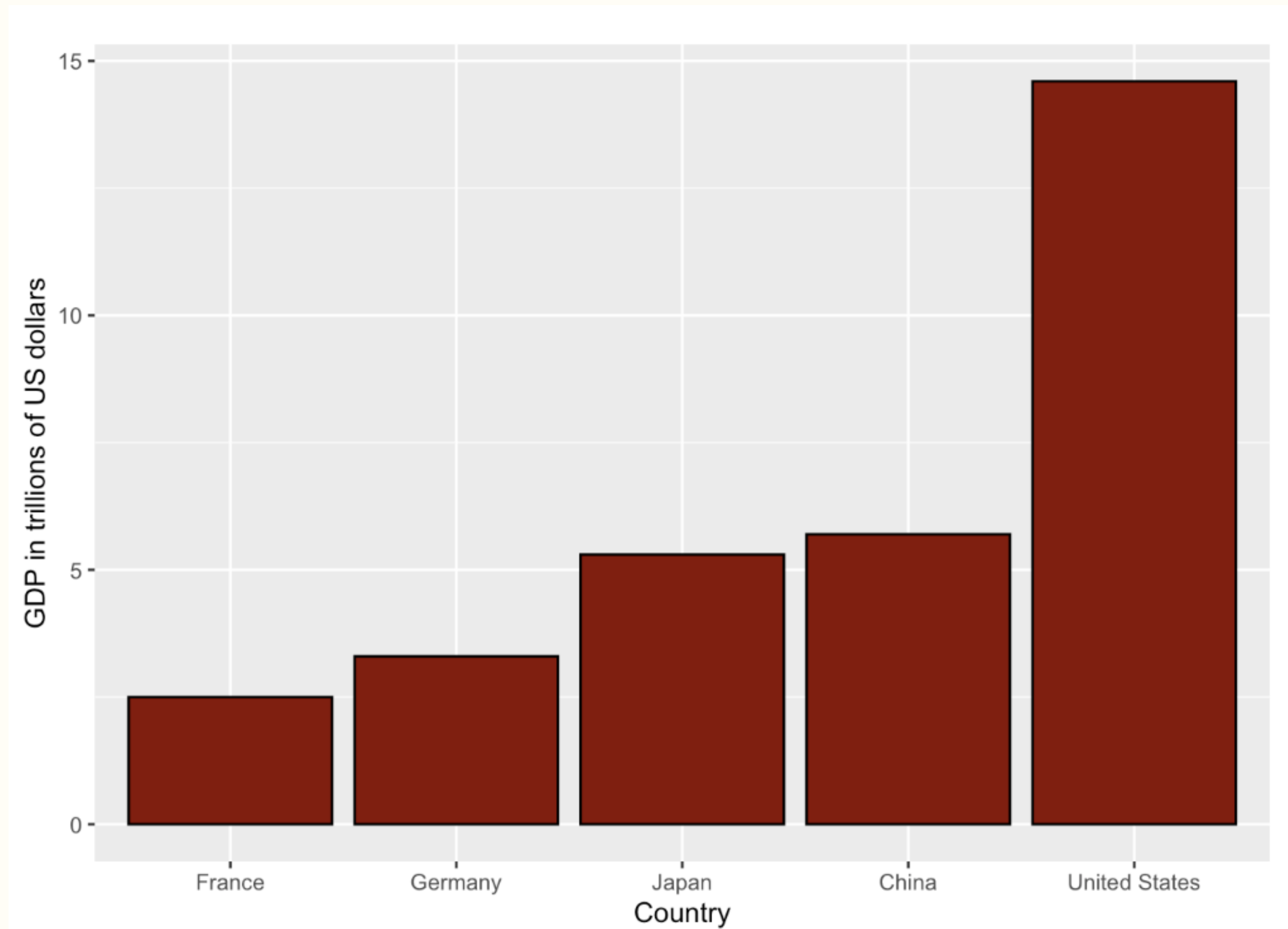
- The US appears to have an economy over 5 times larger than China and over 30 times larger than France
- The actual ratios are 2.6 and 5.8
- The radius, rather than the area, was made to be proportional to the quantity, implying the proportion between the areas is squared
- 2.6 turns into 6.5 and 5.8 turns into 34.1



Distortion of quantities



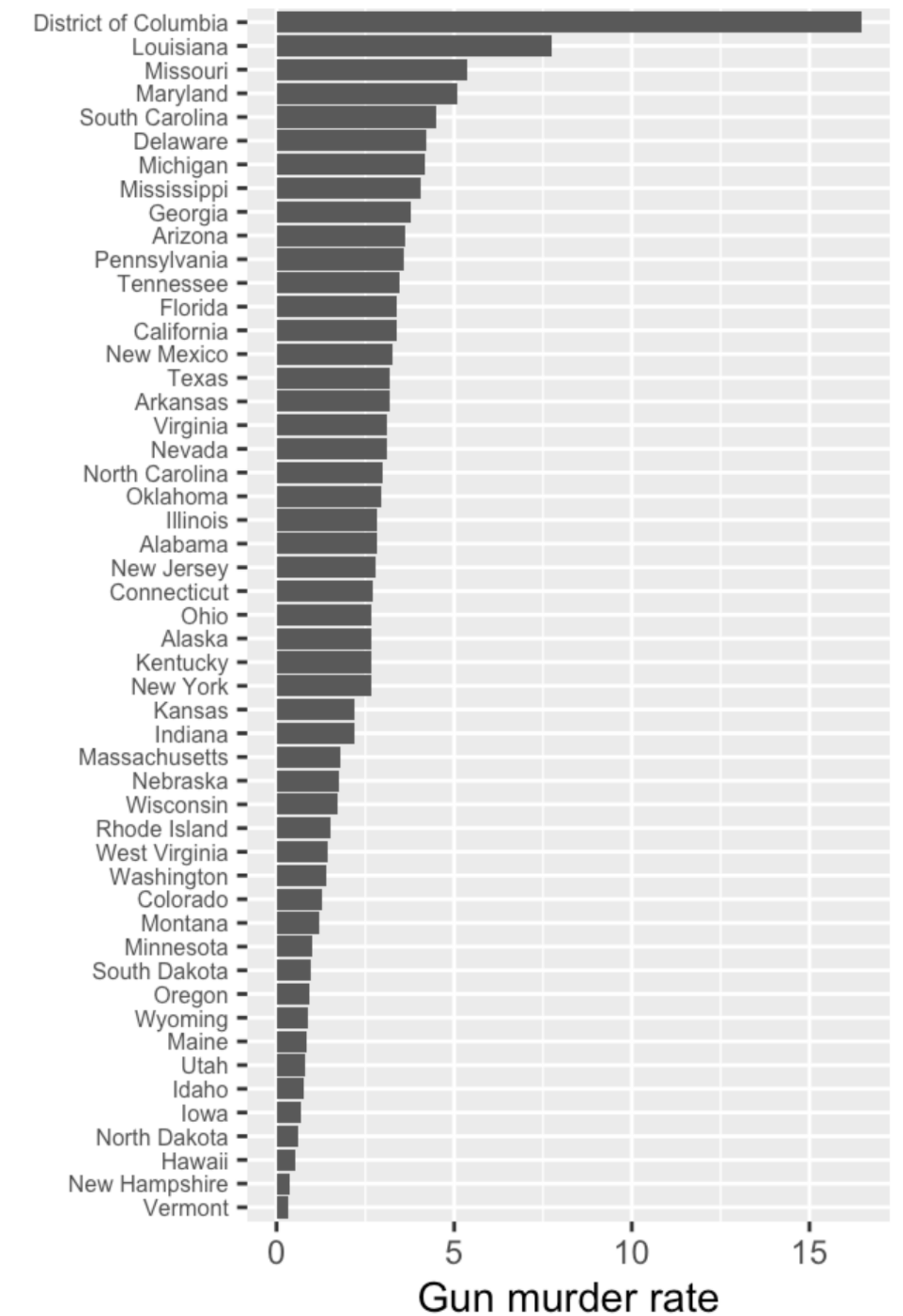
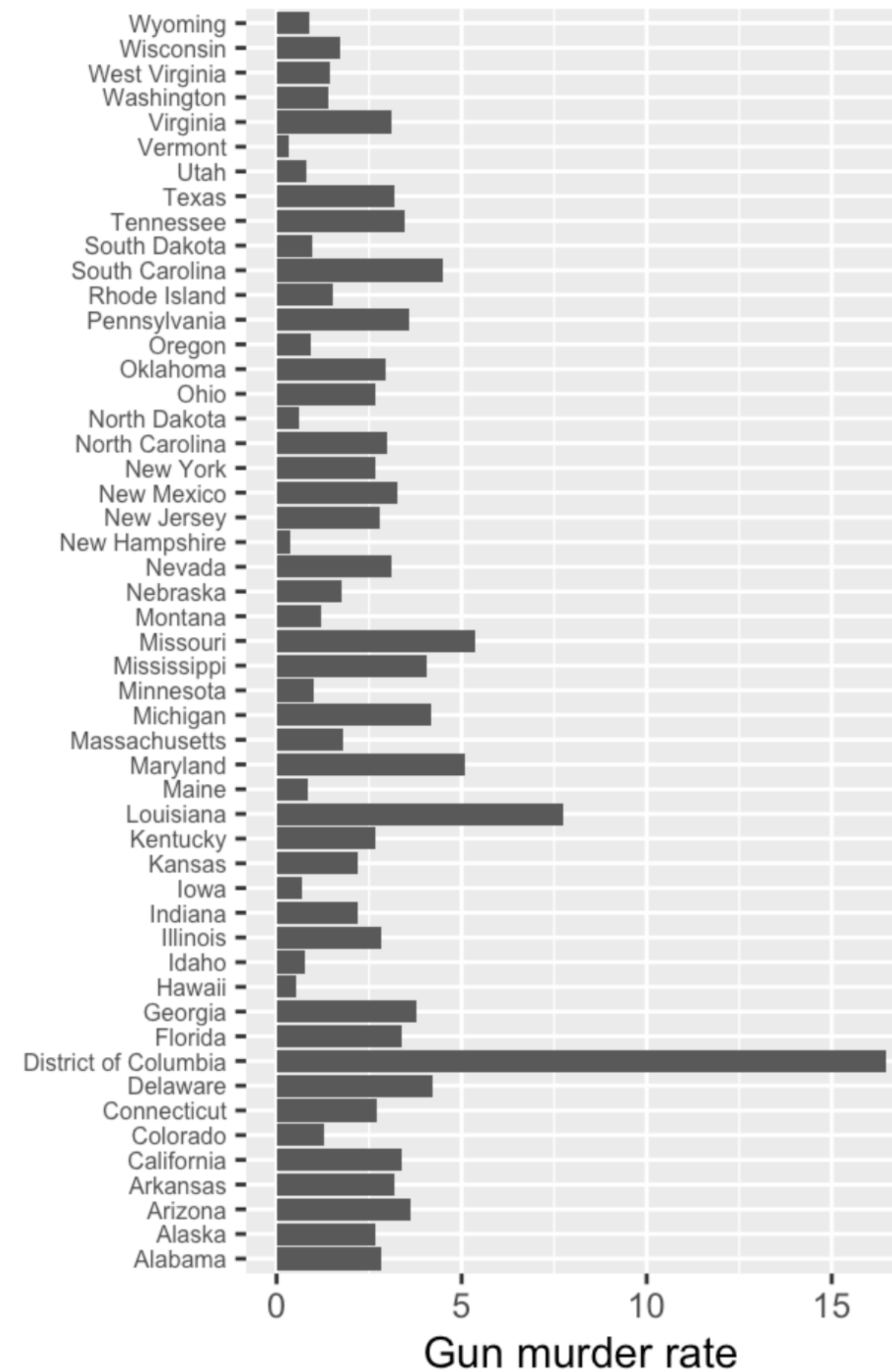
Distortion of quantities



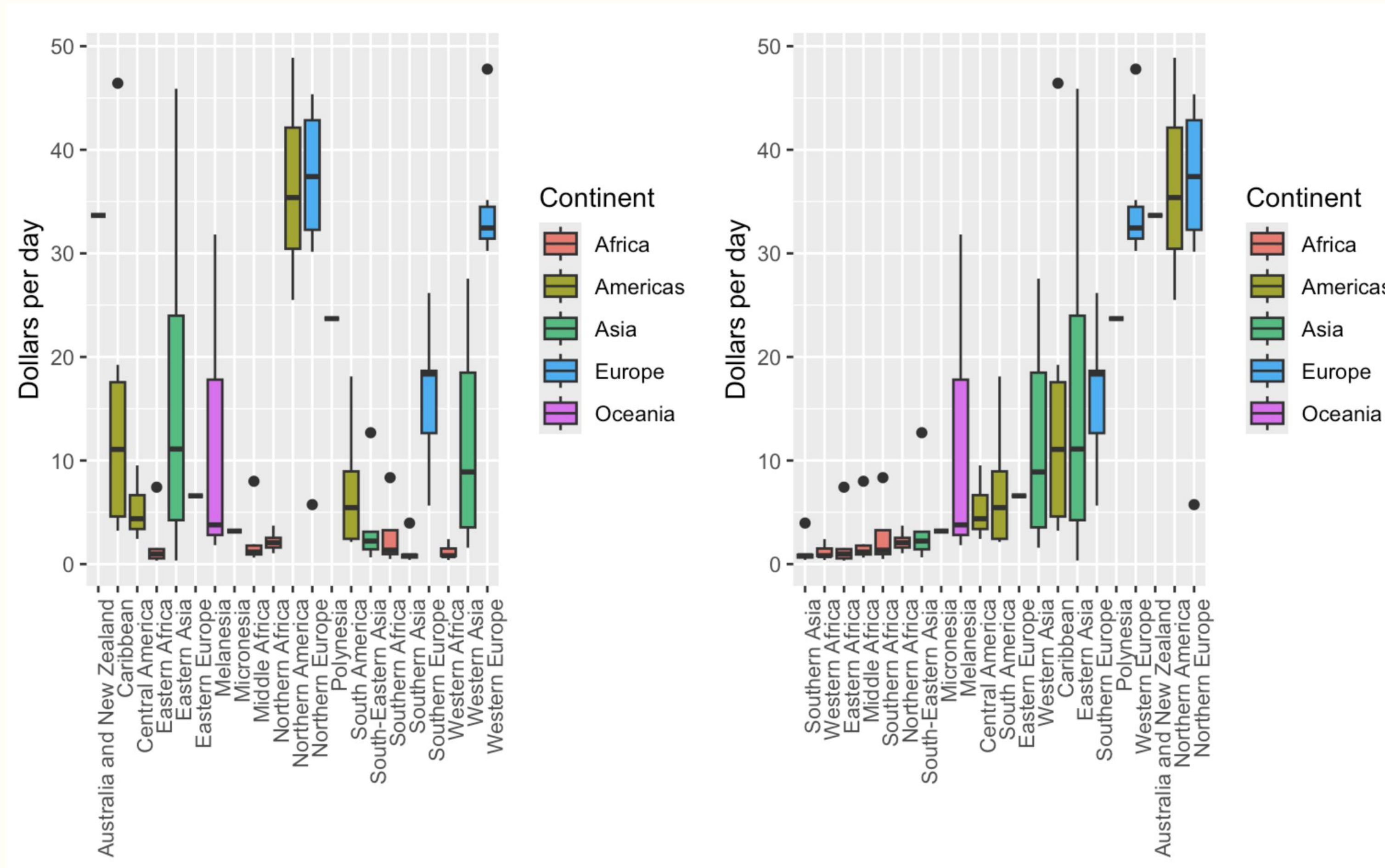
Ordering by a meaningful value

The default **ggplot** behavior is to order categories alphabetically

- Ordering by another value could make visualizations easier to interpret

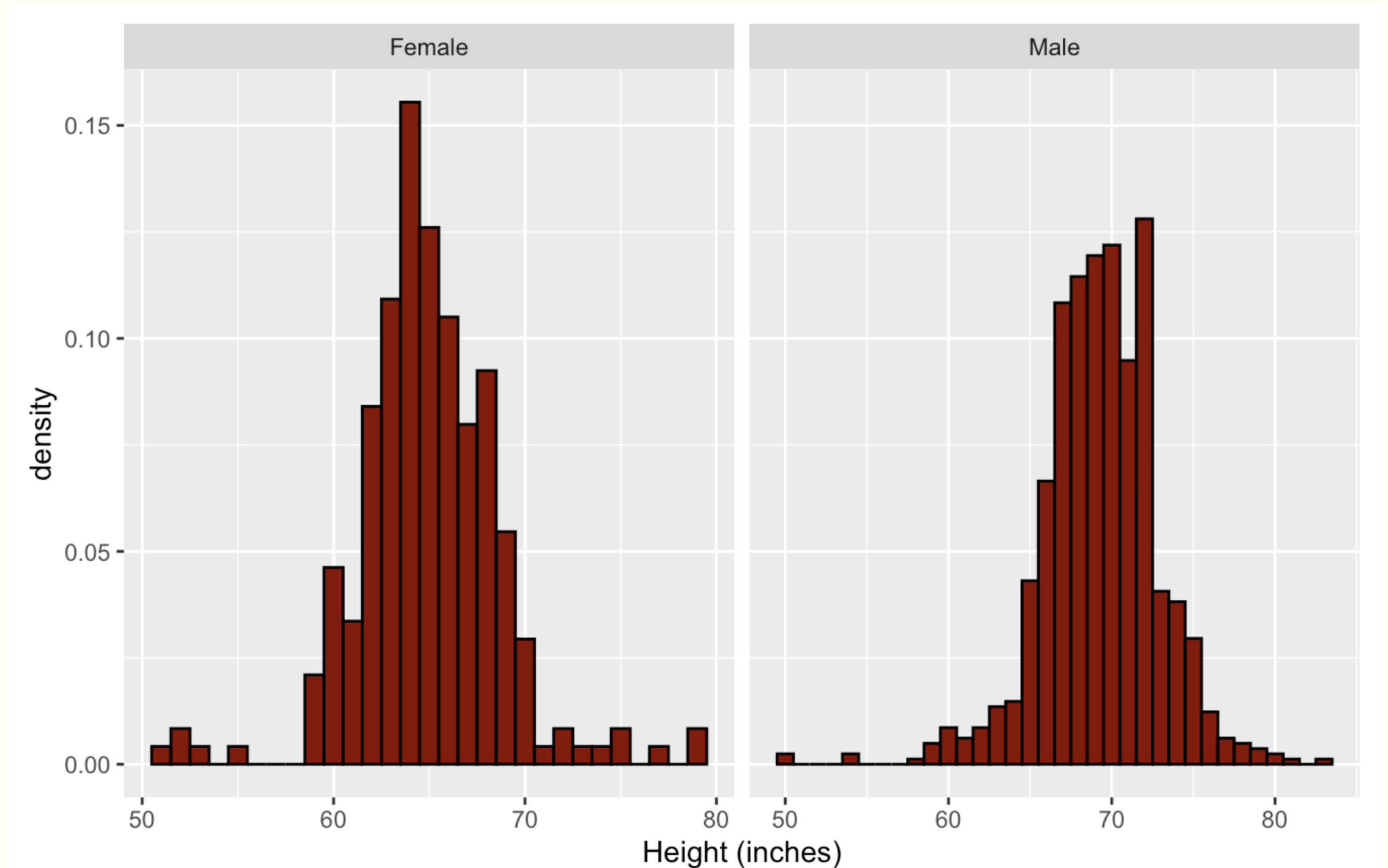


Ordering by a meaningful value



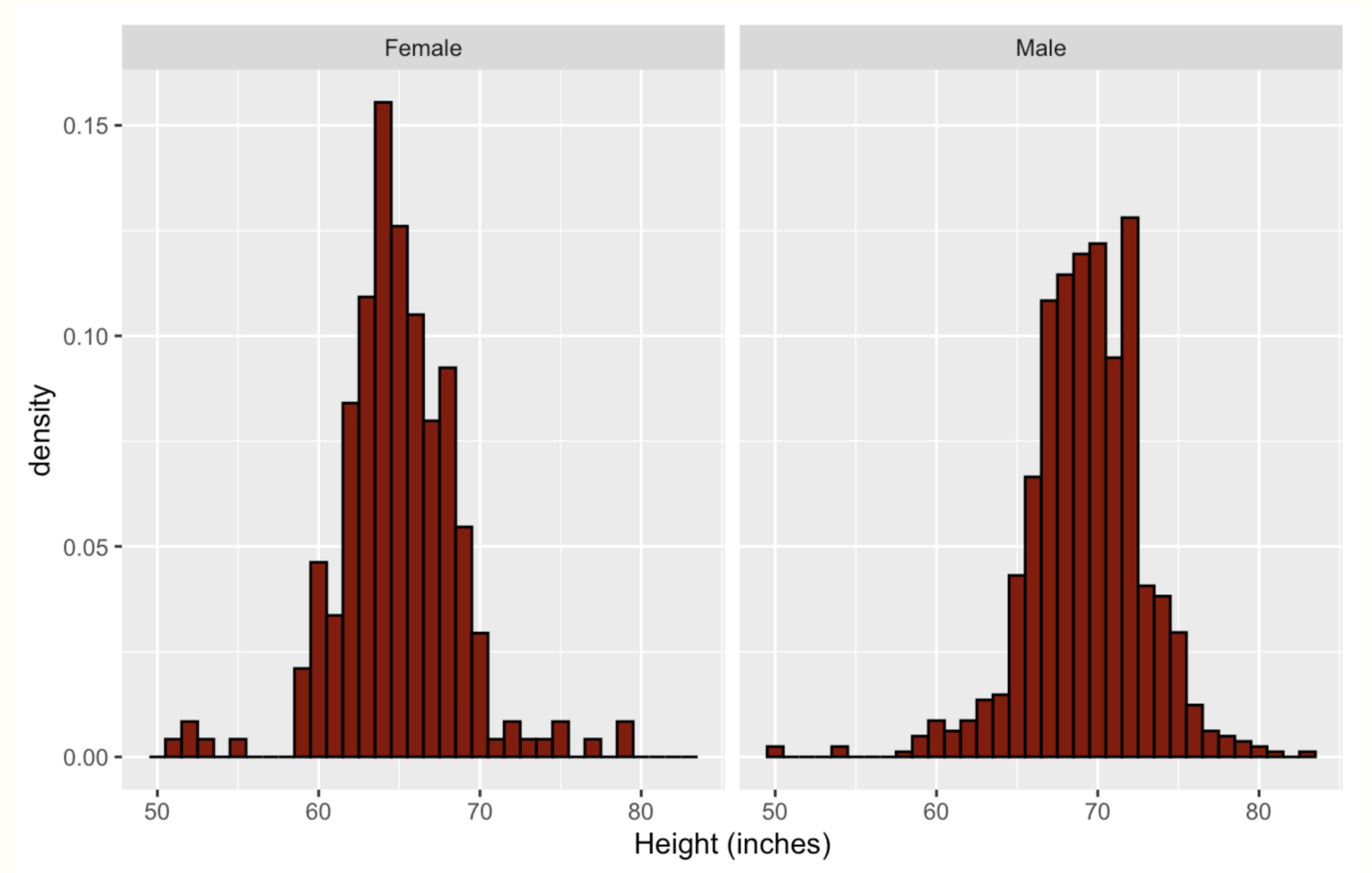
Easing comparisons

- Keep the axes the same when comparing data across plots



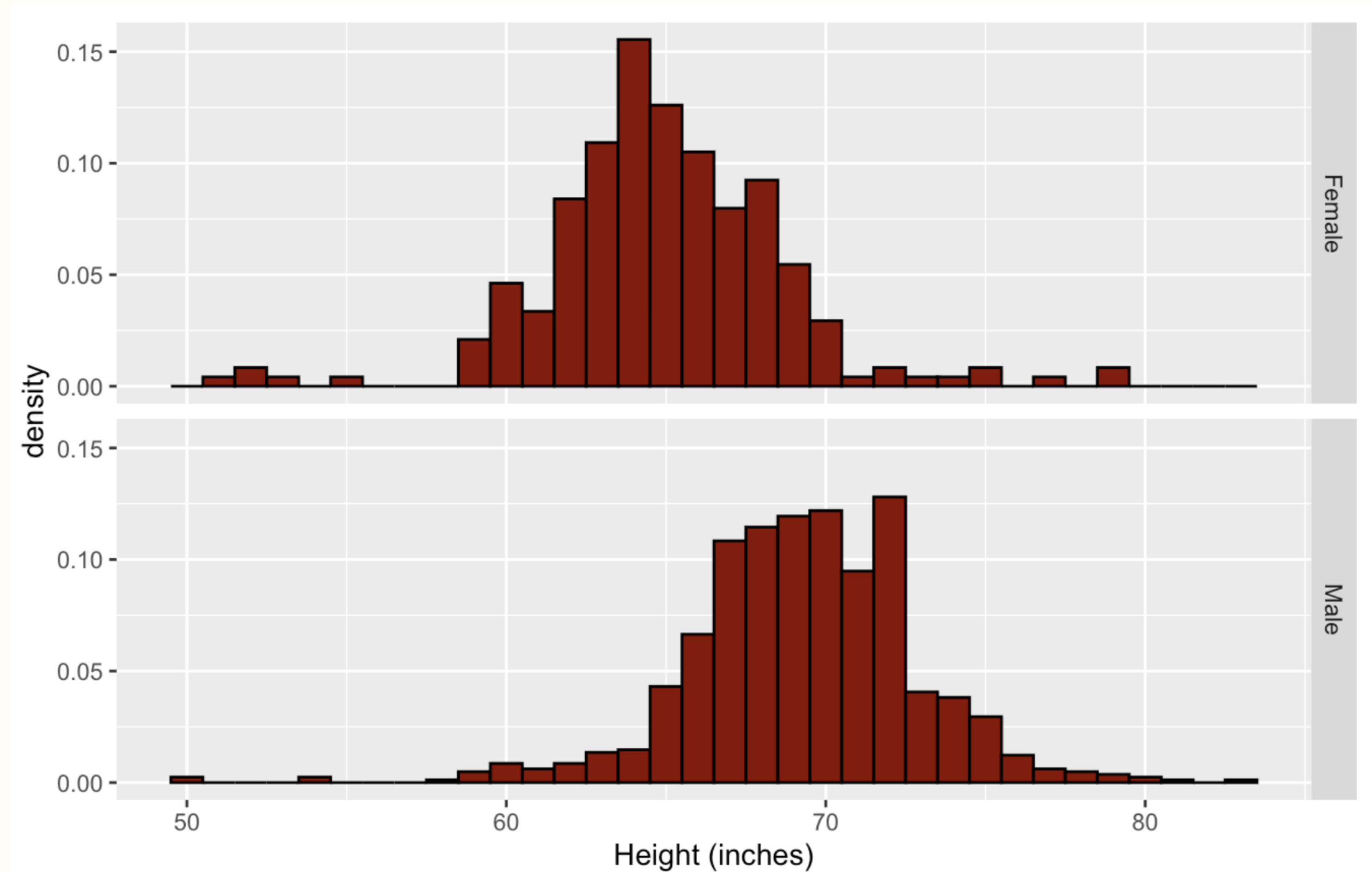
Easing comparisons

- Keep the axes the same when comparing data across plots



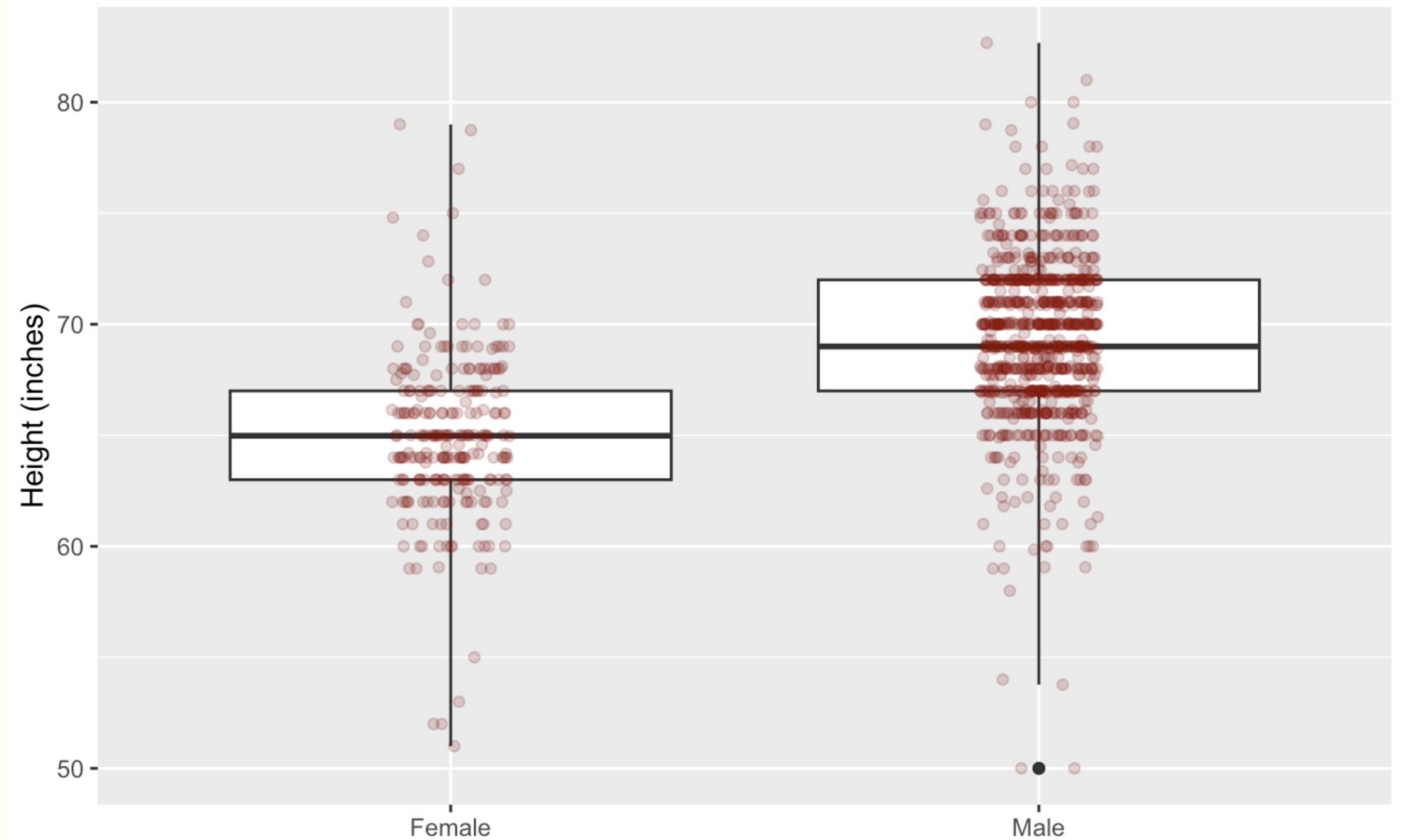
Easing comparisons

- Keep the axes the same when comparing data across plots
- Align plots vertically to see horizontal changes and horizontally to see vertical changes



Easing comparisons

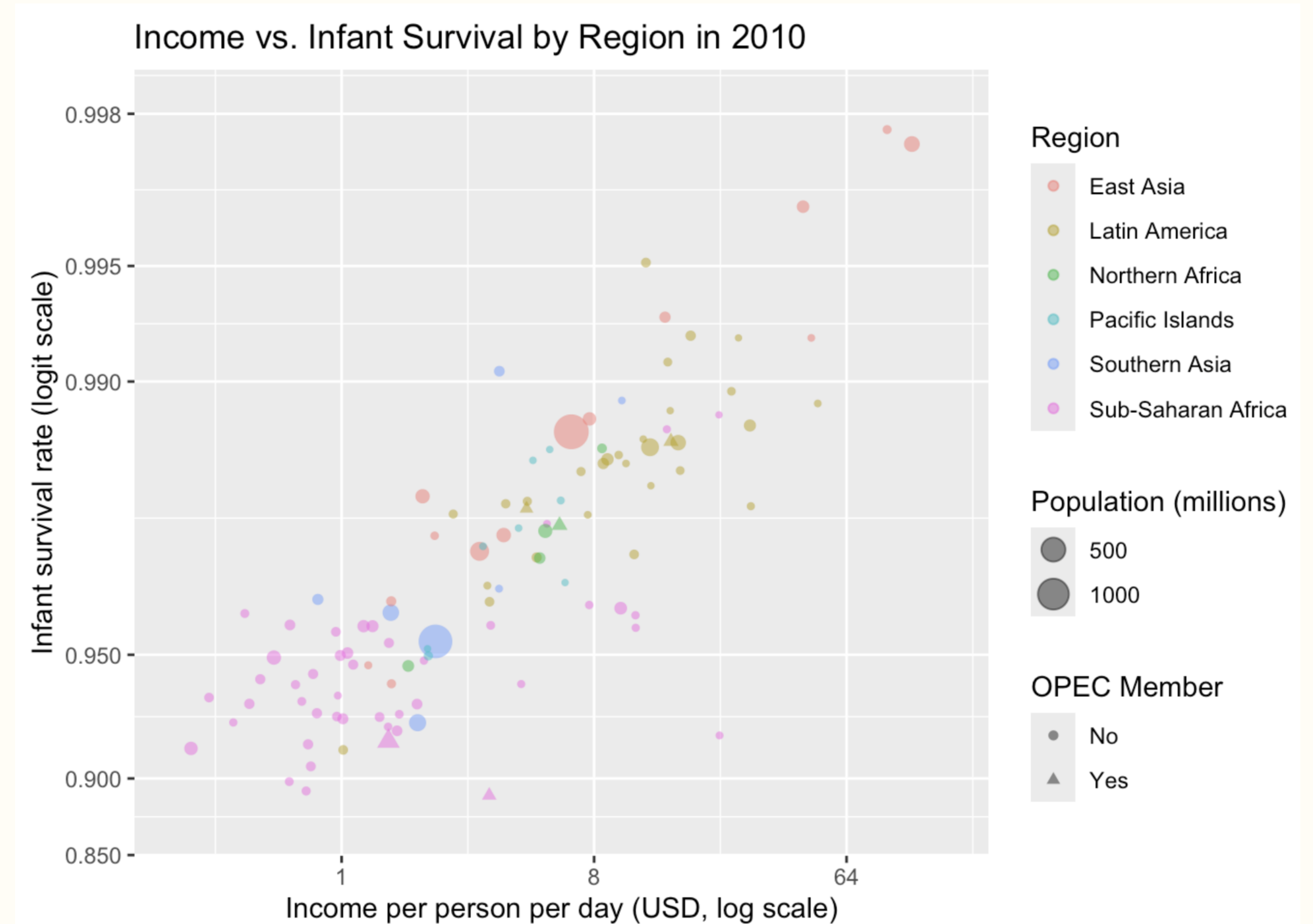
- Keep the axes the same when comparing data across plots
- Align plots vertically to see horizontal changes and horizontally to see vertical changes



Encoding a third variable

Encoding more aspects of a dataset can increase the information provided by the visualization

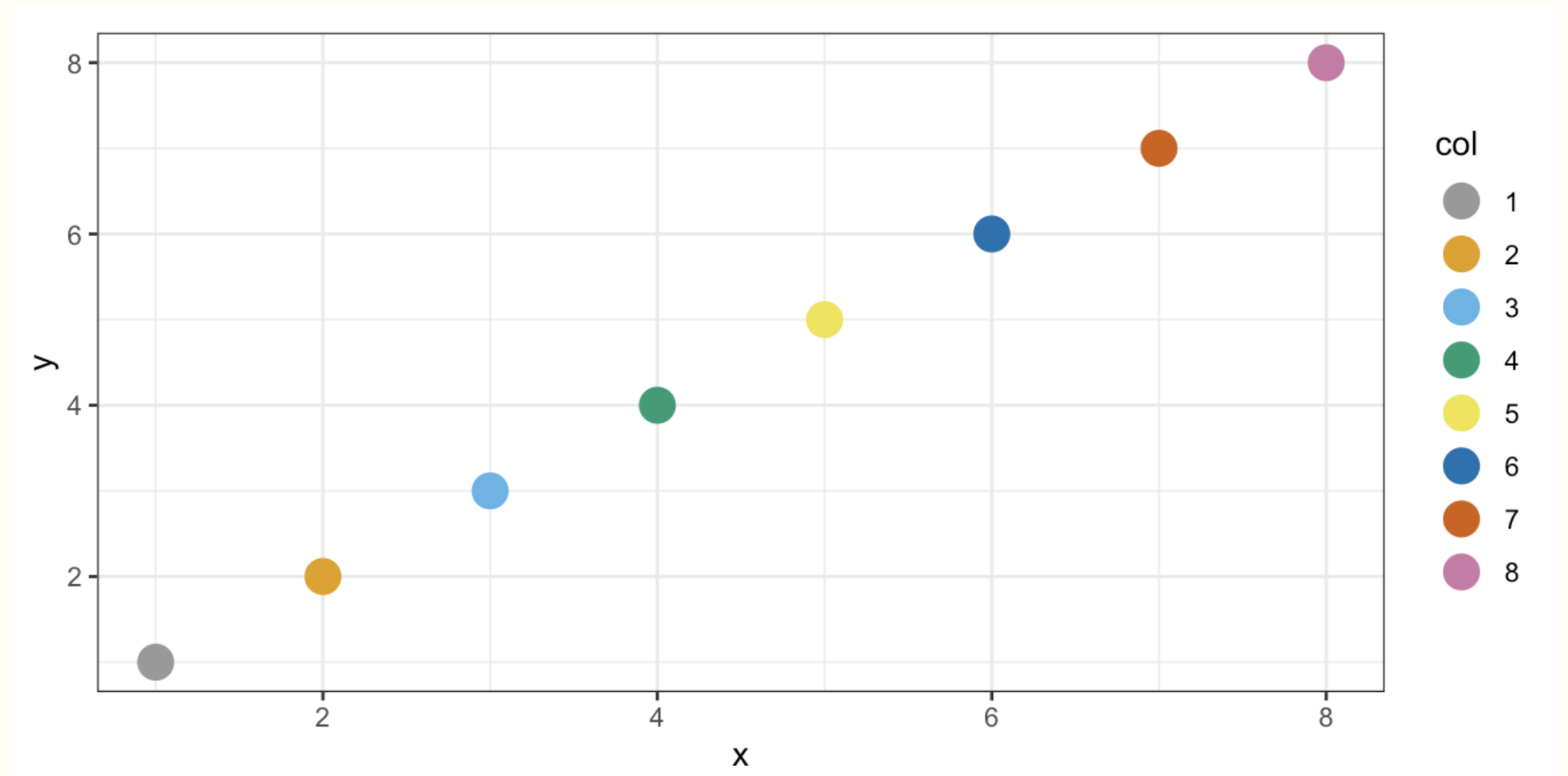
- Must be done carefully – don't clutter the visualization



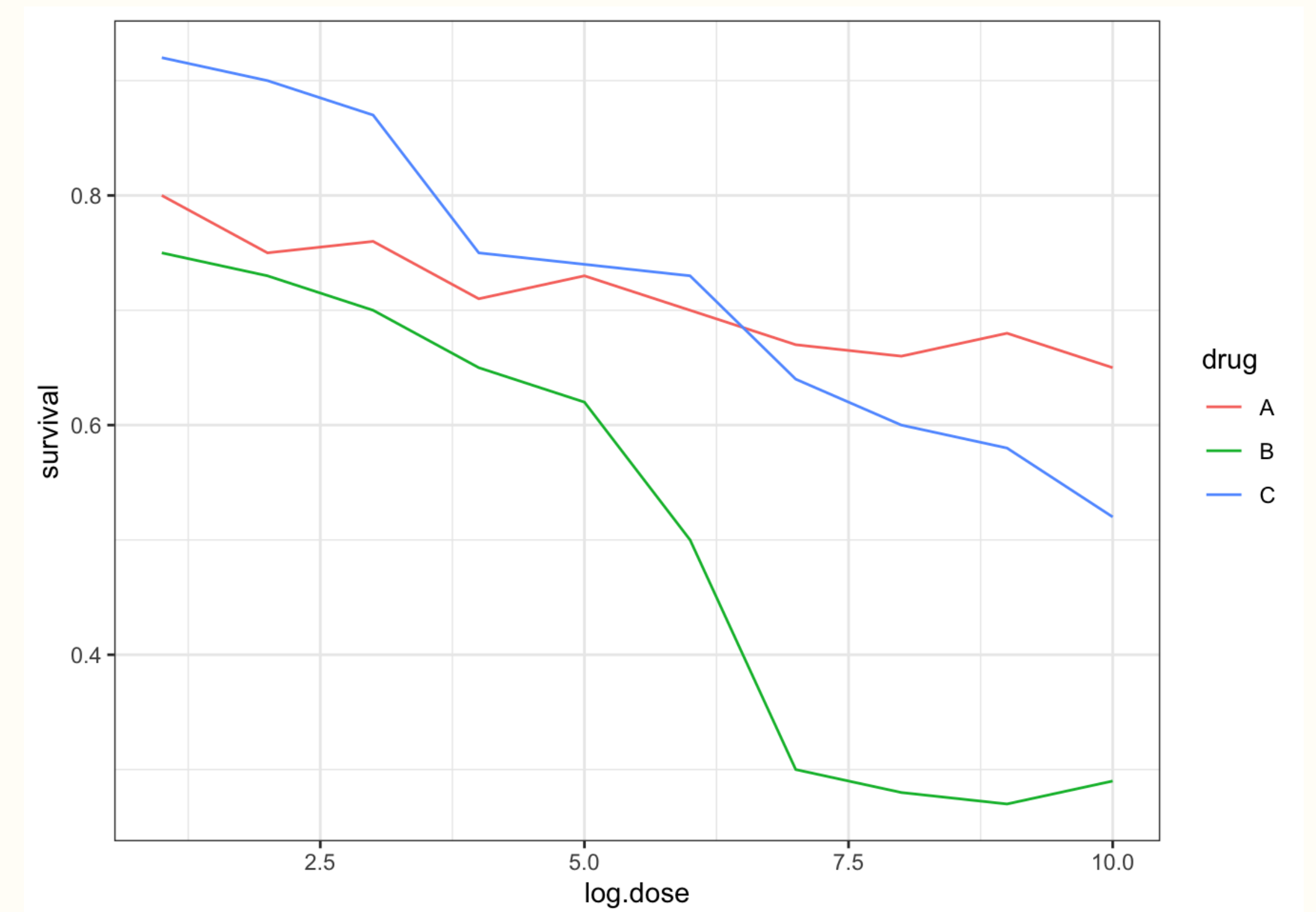
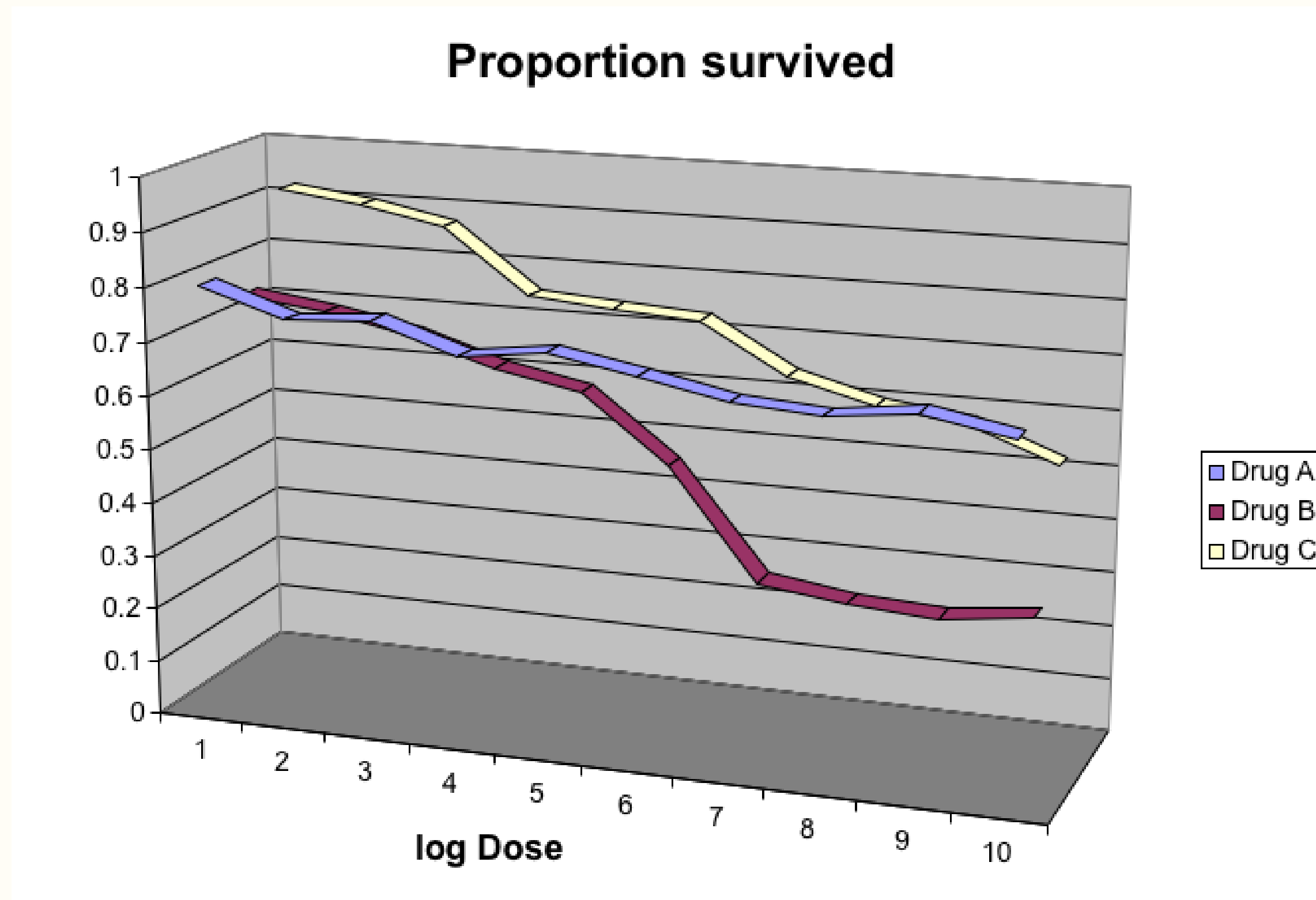
Accessible color palettes

At least 10% of the population is **colorblind**

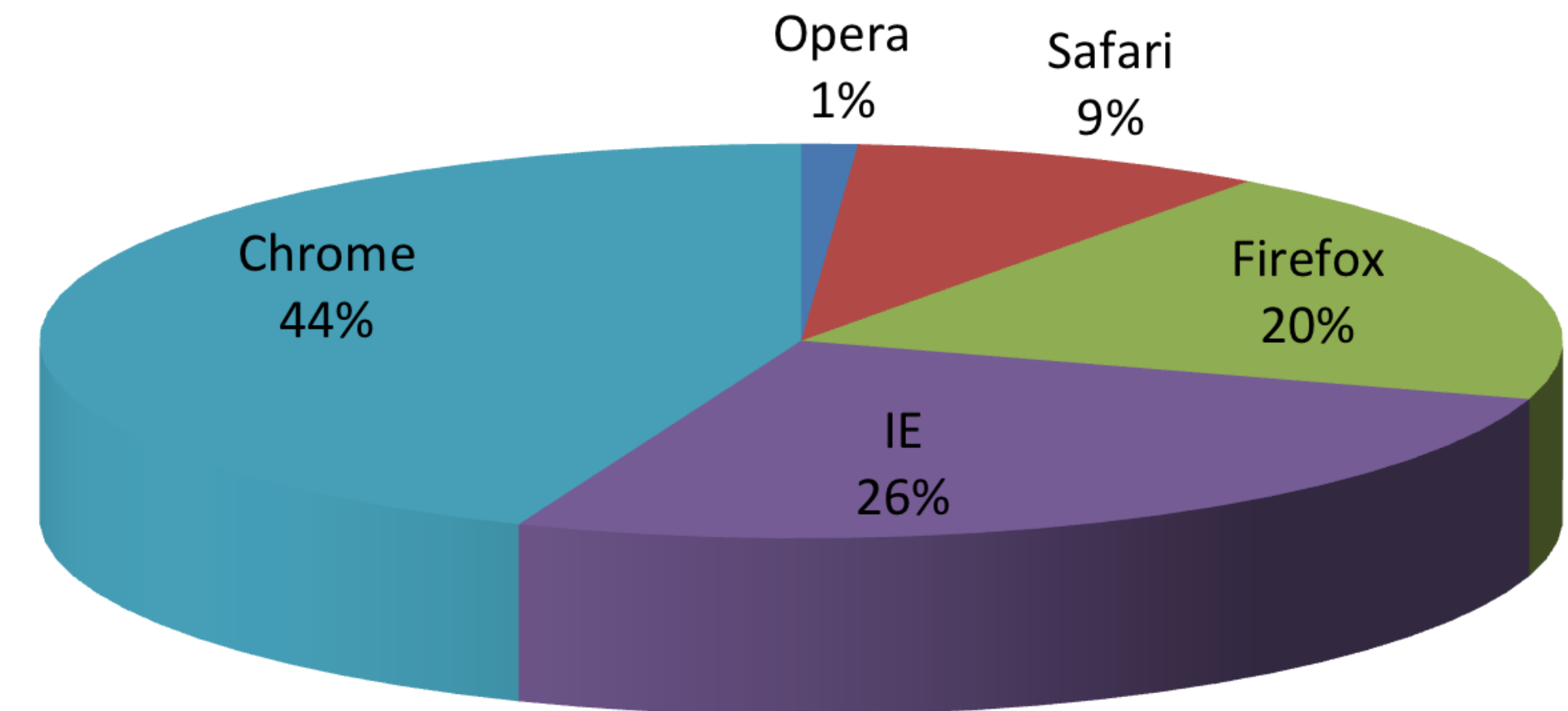
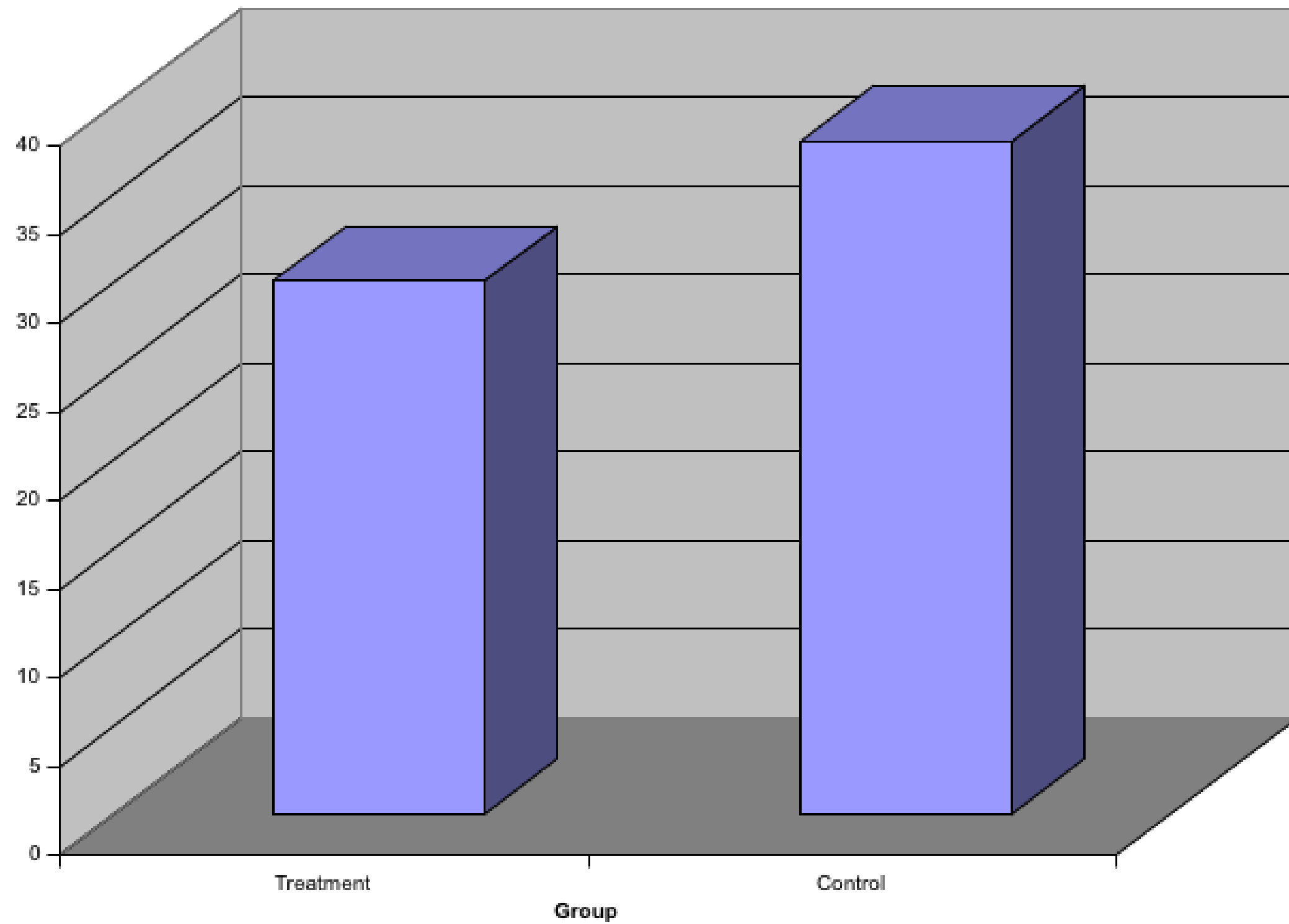
- The default colors used in **ggplot** plots are not colorblind-friendly
- There are palettes that can be used instead



Pseudo 3D plots



Pseudo 3D plots



Significant digits

The default in R is to show 7 decimal places

- Too many significant digits adds visual clutter, making the visualization more difficult to interpret
- Too many significant digits usually does not add information
- Often, 2 decimal places will suffice

state	year	Measles	Pertussis	Polio
California	1940	37.8826320	18.3397861	0.8266512
California	1950	13.9124205	4.7467350	1.9742639
California	1960	14.1386471	NA	0.2640419
California	1970	0.9767889	NA	NA
California	1980	0.3743467	0.0515466	NA

state	year	Measles	Pertussis	Polio
California	1940	37.9	18.3	0.8
California	1950	13.9	4.7	2.0
California	1960	14.1	NA	0.3
California	1970	1.0	NA	NA
California	1980	0.4	0.1	NA



Audience

Know your **audience**

- 1) Our own exploratory data analysis
- 2) To convey a message to experts
- 3) To help tell a story to a general audience



Make sure the intended audience understands each element of the plot

Summary

- Good visualizations prioritize clarity, accuracy, and accessibility
- Humans are much better at processing length rather than angles or area
- The intended audience usually dictates the way data is presented

Introduction to data visualization

Case study: vaccines

Heather Mattie, PhD

Vaccines

Vaccines have helped save millions of lives

However, today, despite all the scientific evidence for their importance, vaccination programs have become somewhat controversial, and an anti-vaccination mentality is on the rise

U.S. Cases in 2025

Total cases

1309

Age

Under 5 years: **375 (29%)**

5-19 years: **482 (37%)**

20+ years: **440 (34%)**

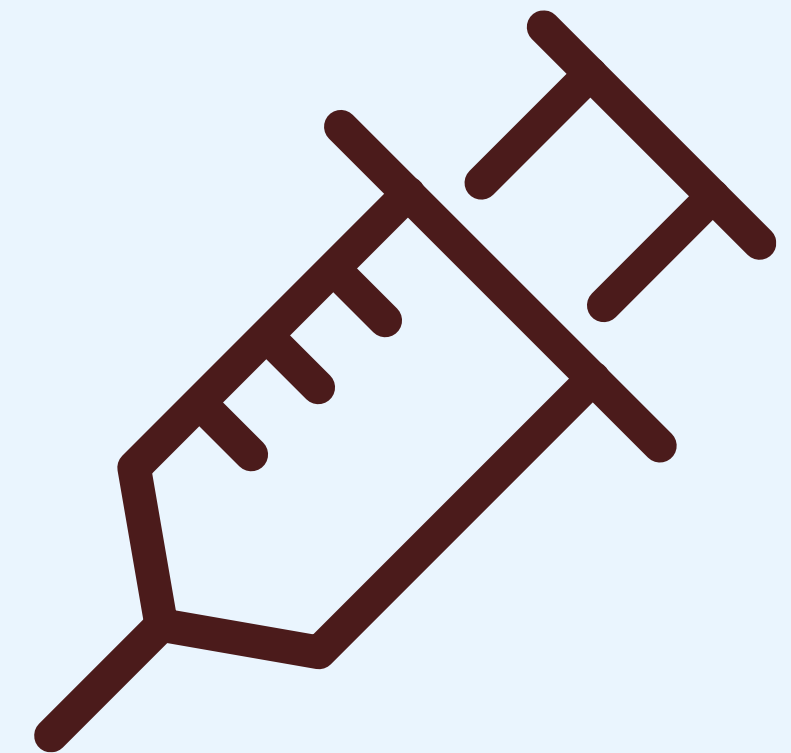
Age unknown: **12 (1%)**

Vaccination Status

Unvaccinated or Unknown: **92%**

One MMR dose: **4%**

Two MMR doses: **4%**



us_contagious_diseases

The **us_contagious_diseases** dataset contains yearly counts of cases of Hepatitis A, Measles, Mumps, Pertussis, Polio, Rubella, and Smallpox for US states

- 1928 – 2011
- Some states have missing data for some years

```
library(tidyverse)
library(dslabs)
data(us_contagious_diseases)
head(us_contagious_diseases)
```

```
##      disease  state year weeks_reporting count population
## 1 Hepatitis A Alabama 1966           50    321    3345787
## 2 Hepatitis A Alabama 1967           49    291    3364130
## 3 Hepatitis A Alabama 1968           52    314    3386068
## 4 Hepatitis A Alabama 1969           49    380    3412450
## 5 Hepatitis A Alabama 1970           51    413    3444165
## 6 Hepatitis A Alabama 1971           51    378    3481798
```

us_contagious_diseases

Task: visualize **Measles** cases in each state for each year. Draw a vertical line indicating when the Measles vaccine was introduced.

- Note that some states reported cases for only a proportion of weeks during the year

```
library(tidyverse)
library(dslabs)
data(us_contagious_diseases)
head(us_contagious_diseases)
```

##		disease	state	year	weeks_reporting	count	population
##	1	Hepatitis A	Alabama	1966	50	321	3345787
##	2	Hepatitis A	Alabama	1967	49	291	3364130
##	3	Hepatitis A	Alabama	1968	52	314	3386068
##	4	Hepatitis A	Alabama	1969	49	380	3412450
##	5	Hepatitis A	Alabama	1970	51	413	3444165
##	6	Hepatitis A	Alabama	1971	51	378	3481798

us_contagious_diseases

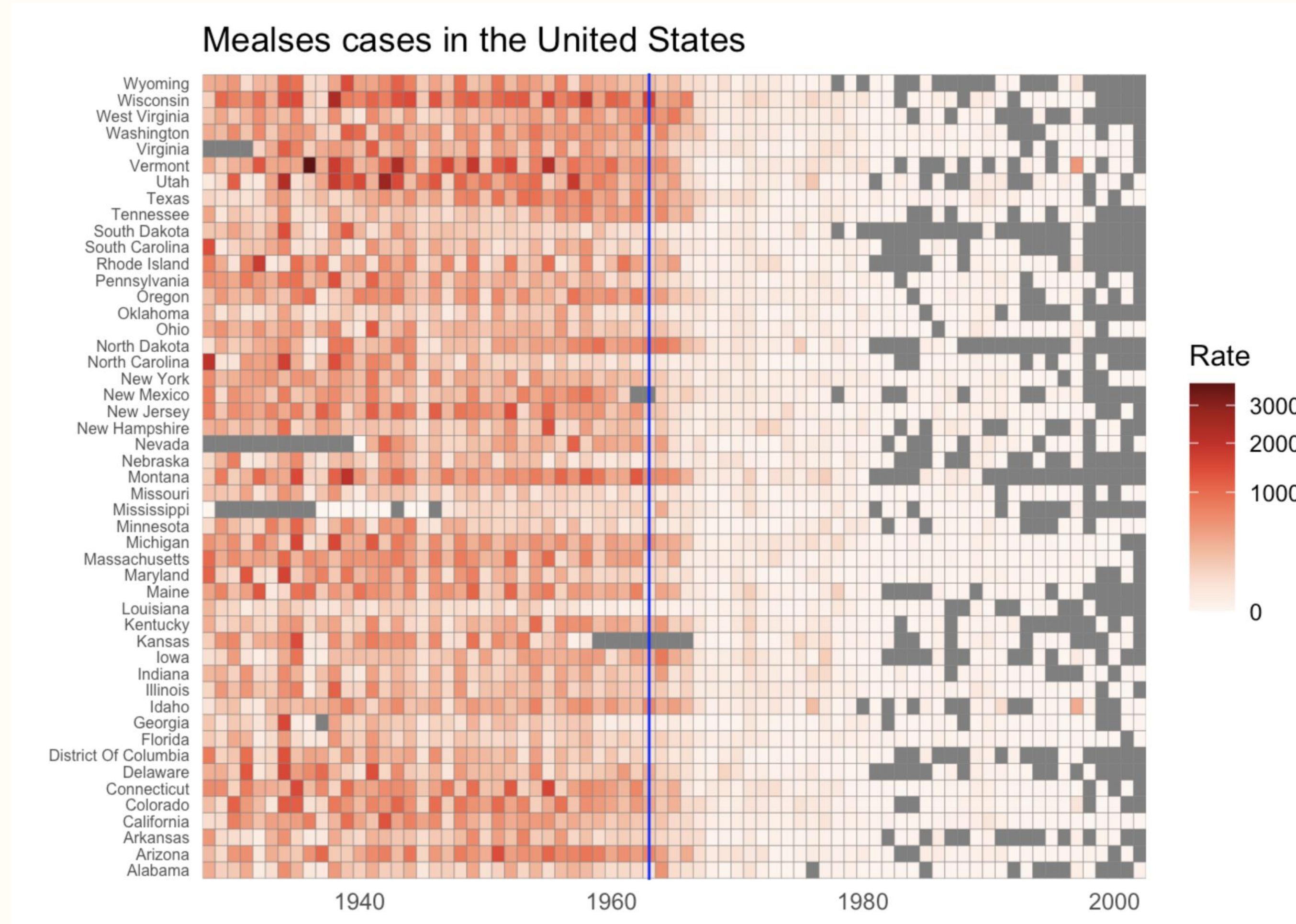
Plan:

1. Create a Measles rate variable taking the **weeks_reporting** variable into account
2. Remove Alaska and Hawaii
3. Create a heatmap

```
library(tidyverse)
library(dslabs)
data(us_contagious_diseases)
head(us_contagious_diseases)
```

##		disease	state	year	weeks_reporting	count	population
## 1	Hepatitis A	Alabama	1966	50	321	3345787	
## 2	Hepatitis A	Alabama	1967	49	291	3364130	
## 3	Hepatitis A	Alabama	1968	52	314	3386068	
## 4	Hepatitis A	Alabama	1969	49	380	3412450	
## 5	Hepatitis A	Alabama	1970	51	413	3444165	
## 6	Hepatitis A	Alabama	1971	51	378	3481798	

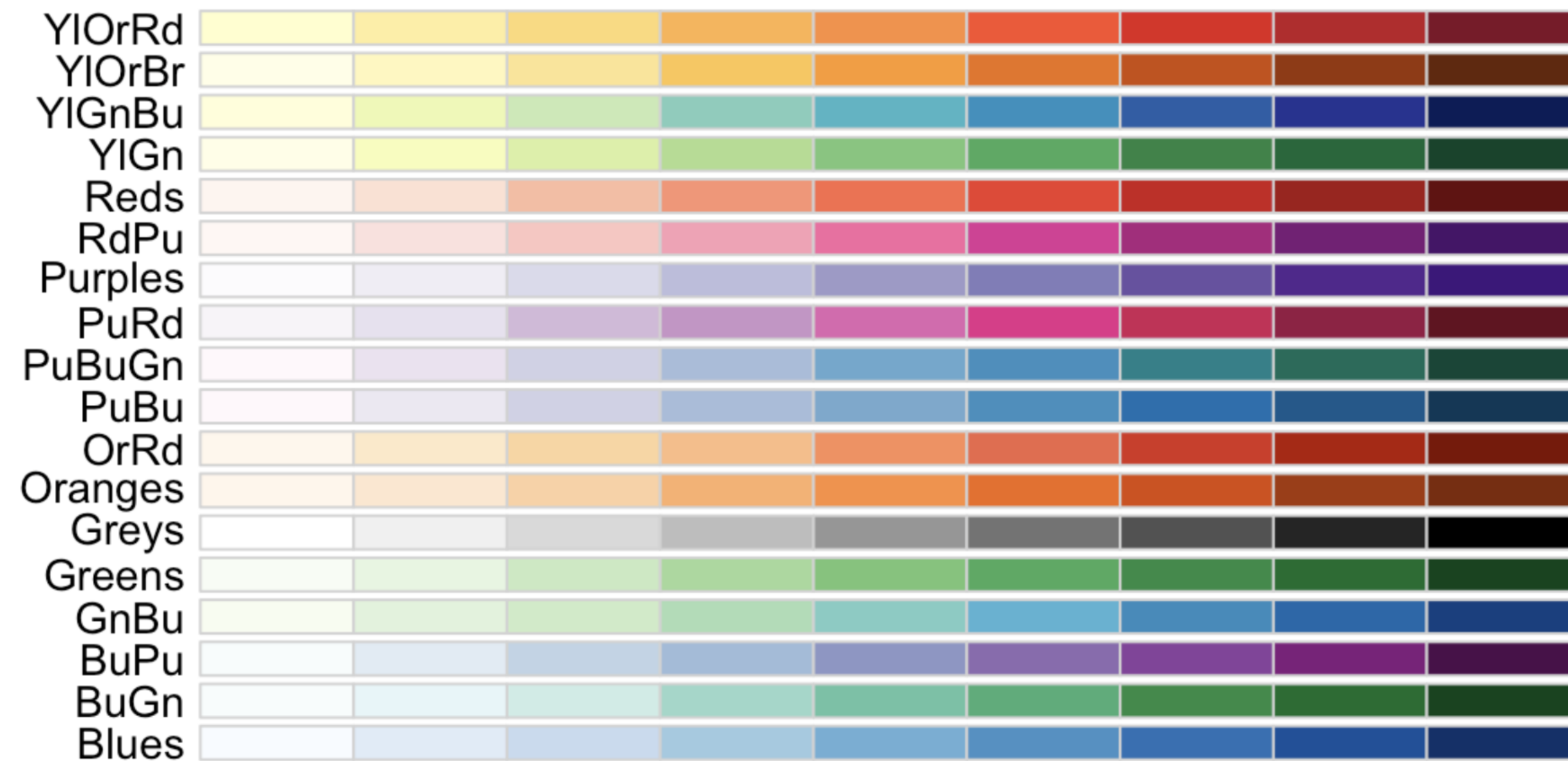
Heatmap



Color palettes

Sequential colors are suited for data with values from high to low

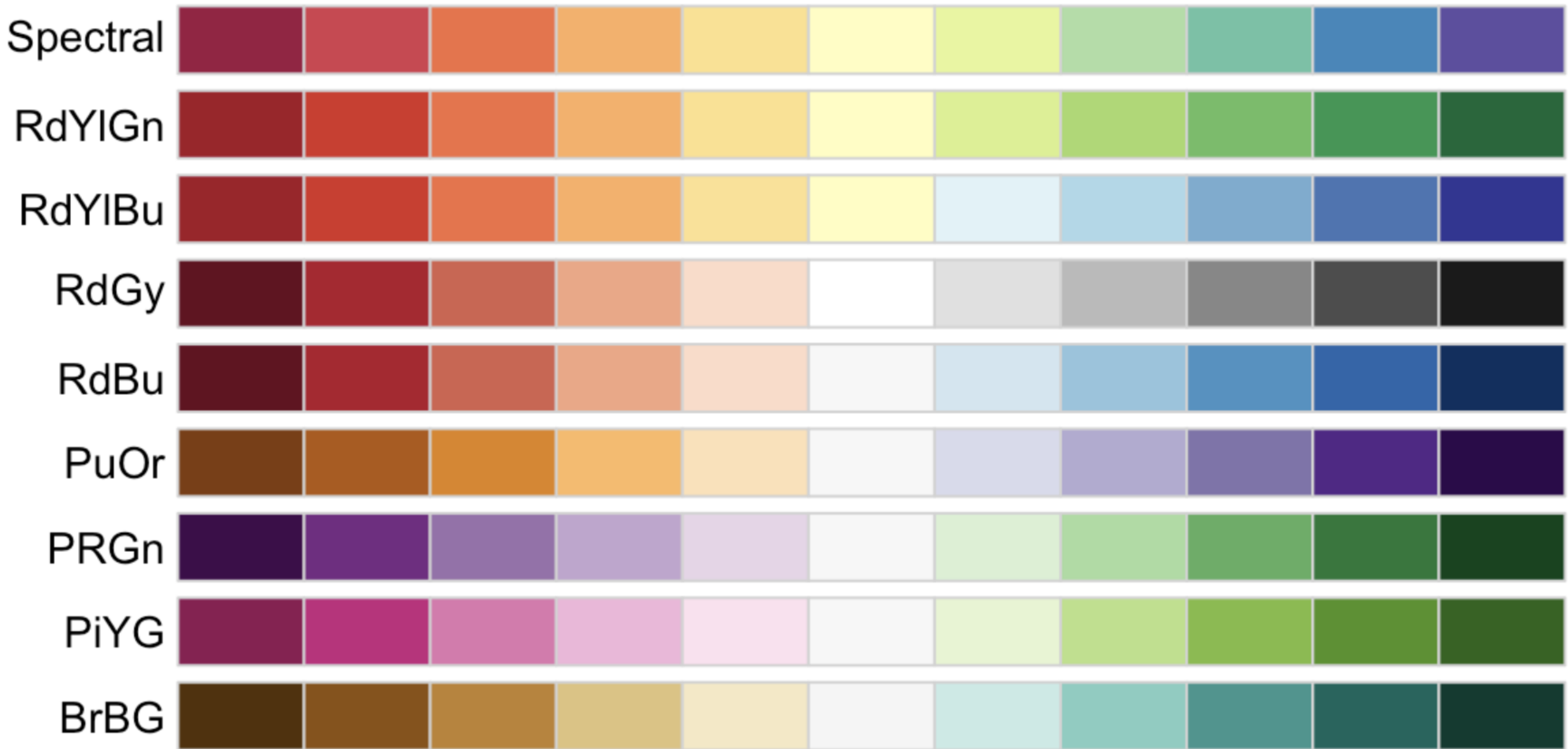
```
library(RColorBrewer)
display.brewer.all(type = "seq")
```



Color palettes

Diverging colors are used to represent values that diverge from a center value

```
display.brewer.all(type="div")
```

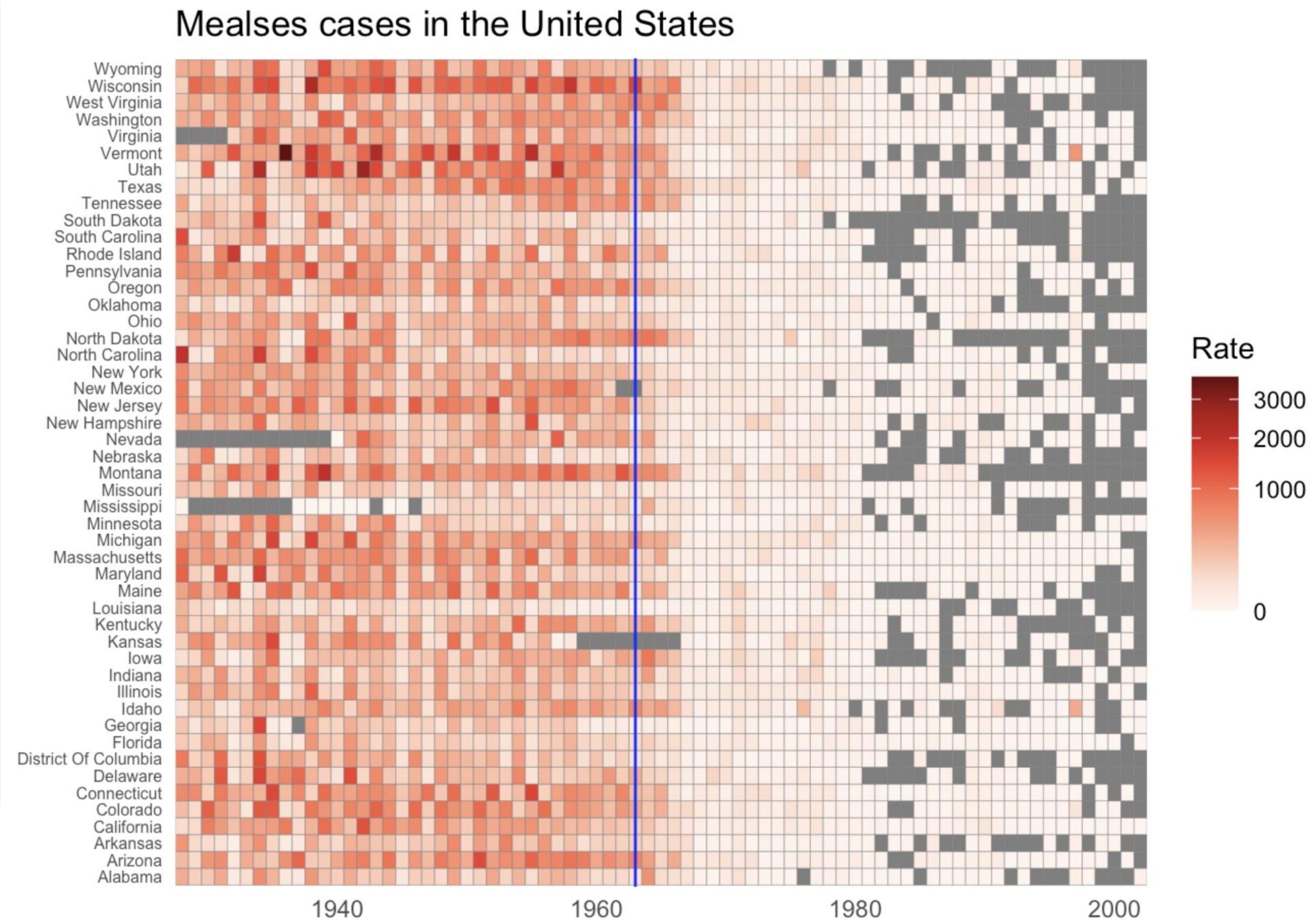


geom_tile

```
library(tidyverse)
library(dslabs)

data <- us_contagious_diseases %>%
  filter(!state %in% c("Hawaii", "Alaska") & disease == "Measles") %>%
  mutate(rate = (count / weeks_reporting) * 52 / (population / 100000))

data %>%
  ggplot(aes(year, state, fill = rate)) +
  geom_tile(color = "grey50") +
  scale_x_continuous(expand = c(0, 0)) +
  scale_fill_gradientn(colors = brewer.pal(9, "Reds"),
    trans = "sqrt",
    name = "Rate") +
  geom_vline(xintercept = 1963, col = "blue") +
  theme_minimal() +
  theme(panel.grid = element_blank(),
    axis.text.y = element_text(size = 6, lineheight = 0.8)) +
  ggtitle("Measles cases in the United States") +
  ylab("") +
  xlab("")
```



Summary

- Data visualizations can be used to make striking arguments in the effectiveness of vaccines and public health in general